WSQ COURSE SLIDES

# Electrostatic Discharge (ESD) Control for Lab and Factory Engineers and Managers

**TGS-2024051412**

TSC: Electrostatic Discharge Control · ELE-OPR-4002-1.1

**Tertiary Infotech Academy Pte Ltd · UEN 201200696W**

Trainer: Dr Alfred Ang

**v3.0 · 19 August 2026**



MANDATORY

# Digital Attendance (TRAQOM · SSG)

1

## AM attendance

Scan the live QR code shown by the trainer.

2

## PM attendance

Complete the required afternoon record.

3

## Assessment attendance

Complete before sitting the assessment.

4

## Use the live LMS

Do not reuse a screenshot or old QR code.


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# About the Trainer




## General Trainer

Complete before class

**NAME** _____

**TITLE / DESIGNATION** _____

**QUALIFICATIONS** _____

**AREAS OF EXPERTISE** _____

**TRAINING & INDUSTRY EXPERIENCE** _____

**CONTACT** _____


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NAMED TRAINER

# About the Trainer




## Dr Alfred Ang

Lead Trainer · Tertiary Infotech Academy

**NAME** Dr Alfred Ang

**TITLE / DESIGNATION** Founder and principal trainer

**QUALIFICATIONS** Doctorate · engineering and digital transformation

**AREAS OF EXPERTISE** Technology, quality, process and AI-enabled work

**TRAINING & INDUSTRY EXPERIENCE** Adult learning and enterprise implementation

**CONTACT** enquiry@tertiaryinfotech.com


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# Let’s know each other

1. Name and role
2. Lab or factory process you support
3. One ESD challenge you want to solve
4. One measurement or control item you already use


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CLASS AGREEMENT

# Ground Rules

1. **Participate**
   Ask, challenge and share evidence.

2. **Respect**
   One conversation; disagree with ideas, not people.

3. **Protect confidentiality**
   Use fictional scenarios; do not expose customer data.

4. **Be punctual**
   Return from breaks on time.

5. **Phones silent**
   Step out quietly for urgent calls.

6. **Safety first**
   Stop any activity that conflicts with site electrical or process safety.


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# Two-day course architecture

1

## Day 1

ESD fundamentals, failure mechanisms and control implementation.

2

## Day 2

Standards, measurement, audit, incident response and management control.

3

## Assessment

1 h WA + 1 h Case Study; open book; submit on LMS.


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# Learning Outcomes

1

## LO1

Apply control measures to manage ESD generation based on ESD principles.

2

## LO2

Evaluate and recommend actions to improve ESD control effectiveness in accordance with ESD standards.

3

## Evidence

Eight realistic activities build audit-ready outputs.

4

## Transfer

Every activity mirrors lab/factory decisions.


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# TSC Knowledge and Abilities

1 **K1**
Introduction to ESD theory and principles, test models and ESD control items.

2 **K2**
Static control with respect to established ESD standards.

3 **K3**
Types of ESD instruments used for checks and measurements.

4 **K4**
Preventive measures to reduce the risk of ESD generation.

5 **A1**
Perform control measures to control the generation of ESD.

6 **A2**
Monitor the effectiveness of employed ESD control measures.

7 **A3**
Recommend follow-up actions to reduce the risk of ESD generation.


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ADMIN

# Briefing for Assessment

## CANDIDATE CONDUCT

- Individual work; no discussion.
- Open book: course slides, Learner Guide and approved materials.
- Phones and other materials under the table.
- Use black/blue pen; no correction fluid.

## INSTRUMENTS

- WA: 4 open-ended questions, K1–K4, 1 hour.
- Case Study: 2 open-ended tasks, A1–A3, 1 hour.
- Submit completed answers on the LMS.
- Reassessment and appeal rights apply.


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ADMIN

# Assessment and Funding

## ELIGIBILITY

- Minimum 75% SSG digital attendance.
- Complete both assessment instruments.
- Competent outcome required for successful completion.

## OUTCOME

- Assessment evidence must be your own work.
- Assessor feedback explains the decision.
- Appeal may be lodged through the published process.


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# Assessment Flow

1. TRAQOM digital attendance
2. Assessment digital attendance
3. Sit WA then Case Study
4. Submit answers on the LMS
5. Sign Assessment Summary Record


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# Access the Course Activities

1. Open the course GitHub repository
2. Choose Clone or Download ZIP
3. Open the activities/ folder
4. Start with the numbered README

[Open the course activity repository]


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REASONING CHECKPOINT 1 OF 3

# How to reason about ESD exposure

1

## Recognise the charge source, field and vulnerable item before selecting a control.

Detailed execution steps and evidence templates are in the Learner Guide and activity folder.


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01

MODULE 1 · K1 · A1

# Basics of ESD

Charge generation · discharge paths · HBM/MM/CDM · failure modes · business impact

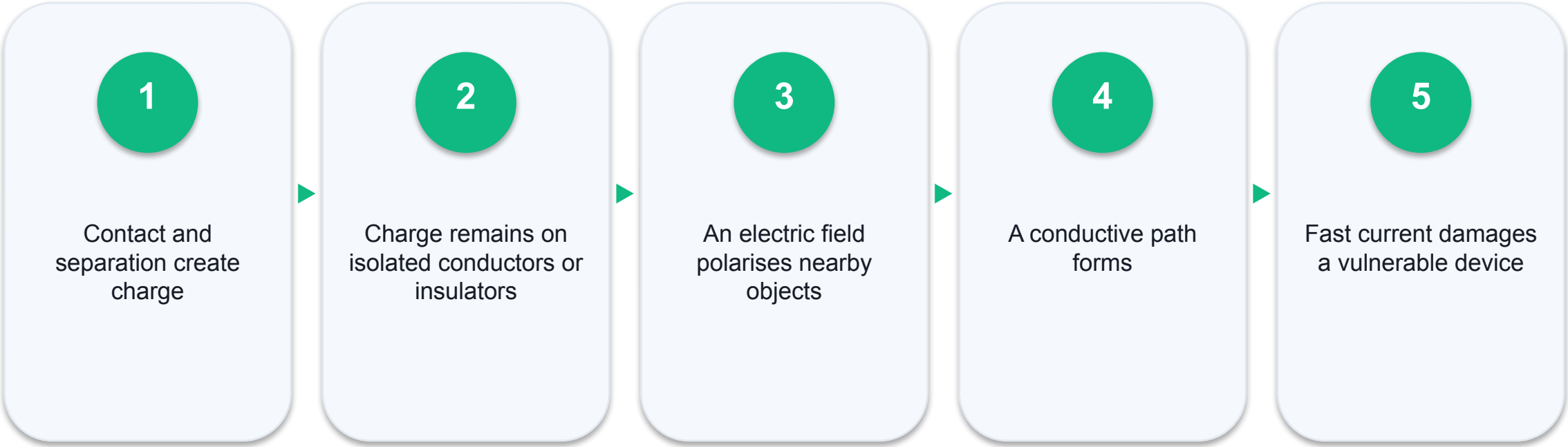

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# A sudden charge transfer can damage a device without a visible spark or a felt shock.

Treat every unprotected ESDS item as vulnerable throughout handling, test, storage and service.


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# The charge–field–discharge chain





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# Triboelectric charging

1

## Material pair

Different electron affinities drive polarity and magnitude.

2

## Contact area

More intimate contact can transfer more charge.

3

## Separation

Speed and force affect charge generation.

4

## Environment

Humidity and contamination change surface leakage.


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# Conductors, dissipative materials and insulators

## PRINCIPLE / USE

- Conductors redistribute charge quickly and can be grounded.
- Dissipative materials move charge in a controlled manner.

## RISK / DECISION

- Insulators retain localised charge and cannot be neutralised by grounding alone.
- Low-charging, dissipative and shielding are different material functions.


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# A metal tool, trolley or fixture can remain charged when it has no verified path to ground.

Bonding removes potential differences; grounding provides a controlled reference path.


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# Electrostatic induction and polarisation

1. Charged object approaches
2. Field redistributes charge in nearby conductor
3. Temporary ground connection permits charge flow
4. Ground is removed
5. Object retains a net charge after the field leaves


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# Contact discharge versus air discharge

## PRINCIPLE / USE

- Contact discharge: differently charged objects touch and charge transfers directly.
- Waveform depends on the source, path impedance and contact geometry.

## RISK / DECISION

- Air discharge: a strong field ionises an air gap before contact.
- Visible sparks represent only a subset of damaging ESD events.


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# What controls discharge severity?

1

## Voltage difference

Stored potential is one risk indicator, not the whole event.

2

## Capacitance

More stored charge can raise available energy.

3

## Path impedance

Resistance and inductance shape peak current and rise time.

4

## Device sensitivity

Pins, package and internal structures determine failure threshold.


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# Human Body Model (HBM)

## PRINCIPLE / USE

- Represents a charged person discharging through a device pin.
- Useful for component qualification and handling robustness.

## RISK / DECISION

- A laboratory model—not a prediction of every factory event.
- Use supplier sensitivity data and the applicable controlled test method.


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# Machine Model (MM): legacy context

## PRINCIPLE / USE

- Historically represented charged metal equipment with a low-impedance discharge.
- Appears in the legacy deck and some older literature.

## RISK / DECISION

- Do not assume correlation with HBM or CDM.
- Use the test model and revision required by the current product/customer specification.


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# Charged Device Model (CDM)

1. Device package becomes charged
2. Device approaches grounded metal
3. One pin makes first contact
4. Very fast discharge leaves the device
5. Local current density stresses internal structures


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# HBM, MM and CDM are not interchangeable

1

## HBM

Charged person → device; handling-oriented component model.

2

## MM

Legacy charged-machine representation; limited correlation.

3

## CDM

Charged device → ground; very fast package-dependent event.


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# Catastrophic failure

1. **Metal melt**
   Current density damages interconnects.

2. **Junction breakdown**
   Semiconductor junction no longer meets specification.

3. **Oxide rupture**
   Thin dielectric is punctured.

4. **Immediate detection**
   Device may fail functional or parametric test.


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# Latent and cumulative degradation

## PRINCIPLE / USE

- A degraded device may still pass initial functional test.
- Leakage or margins can shift after an event.

## RISK / DECISION

- The latent-failure concept is debated; evidence must distinguish ESD from other overstress causes.
- Repeated mishandling can cumulatively degrade parameters.


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# ESD versus electrical overstress (EOS)

## PRINCIPLE / USE

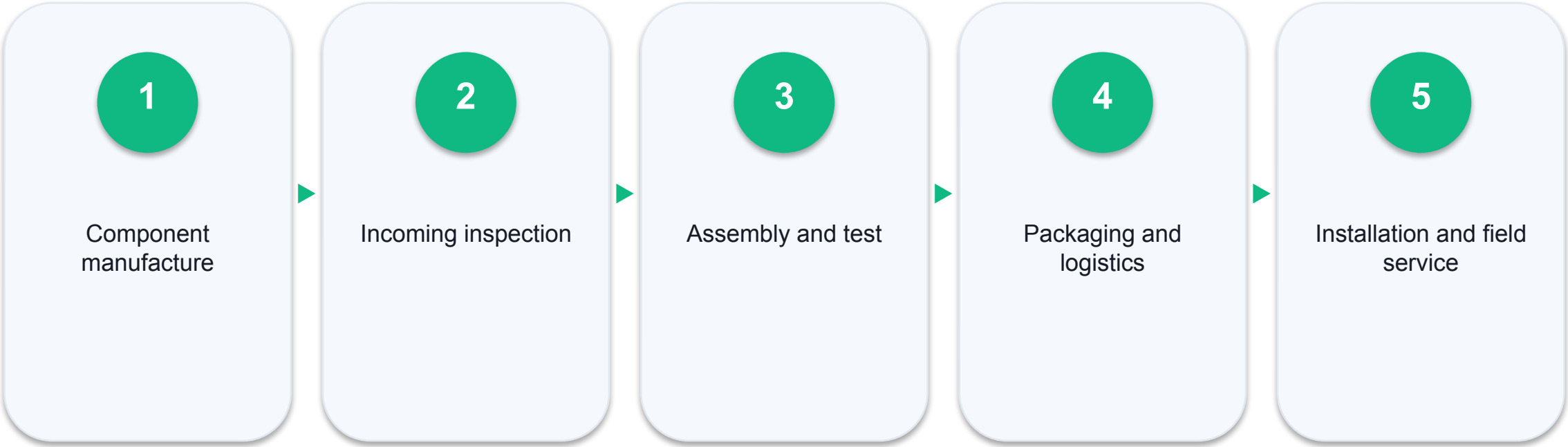
- ESD: rapid electrostatic charge transfer, usually short duration and fast rise time.
- Evidence includes handling history, field sources and microscopic damage pattern.

## RISK / DECISION

- EOS: broader excessive electrical energy exposure from power, wiring or transients.
- Do not label every burnt part as ESD without a disciplined failure analysis.


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# Where exposure occurs





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# Which devices may be sensitive?

1

## Semiconductors

ICs, MOSFETs, diodes and precision analog parts.

2

## Assemblies

Populated PCBs and hybrid circuits.

3

## Sensors

Optoelectronic, microwave, piezoelectric and recording-head devices.

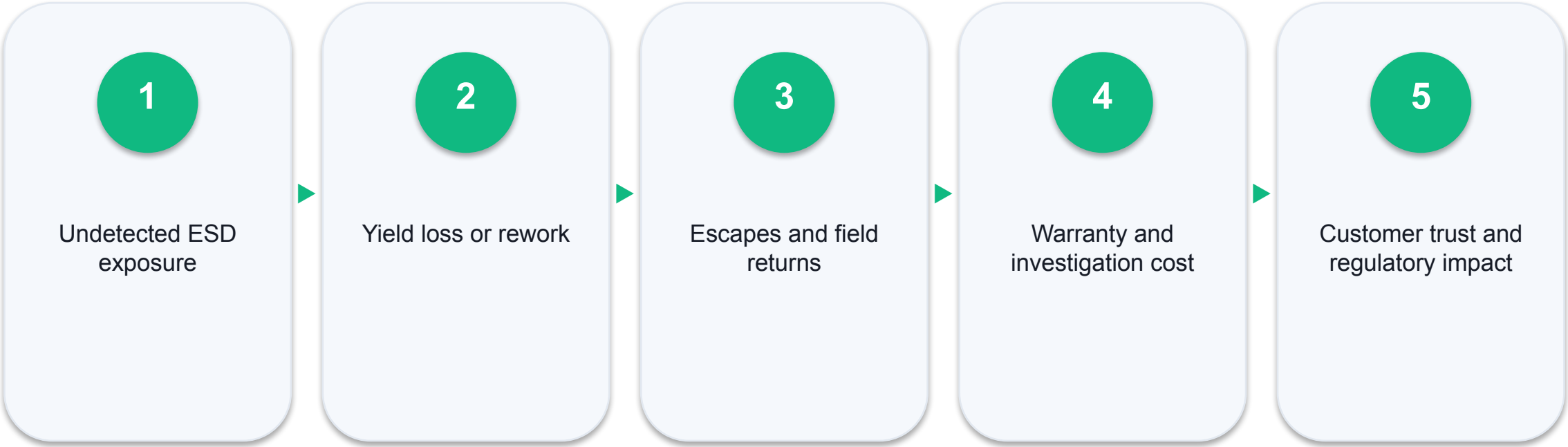
4

## Supplier data

Actual sensitivity comes from product qualification information.


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# Economic impact pathway





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# Risk-based prioritisation

1 **Sensitivity**
Lowest credible device withstand level.

2 **Exposure**
People, motion, insulators, automation and transport.

3 **Detectability**
Whether damage is caught before shipment.

4 **Consequence**
Safety, quality, cost, customer and downtime impact.


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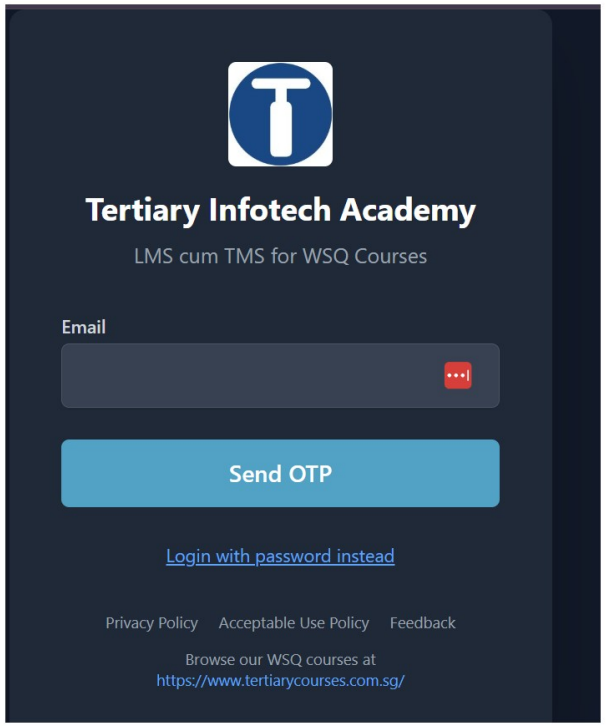


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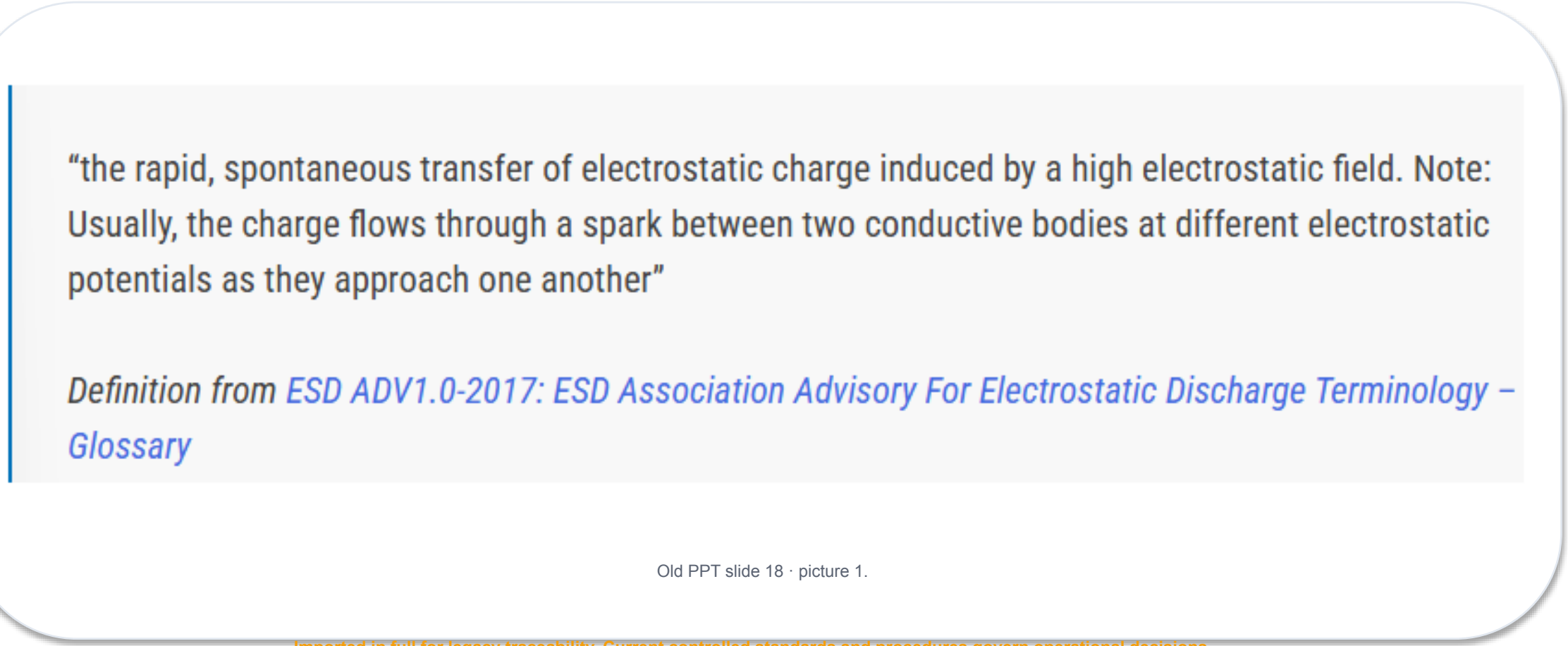



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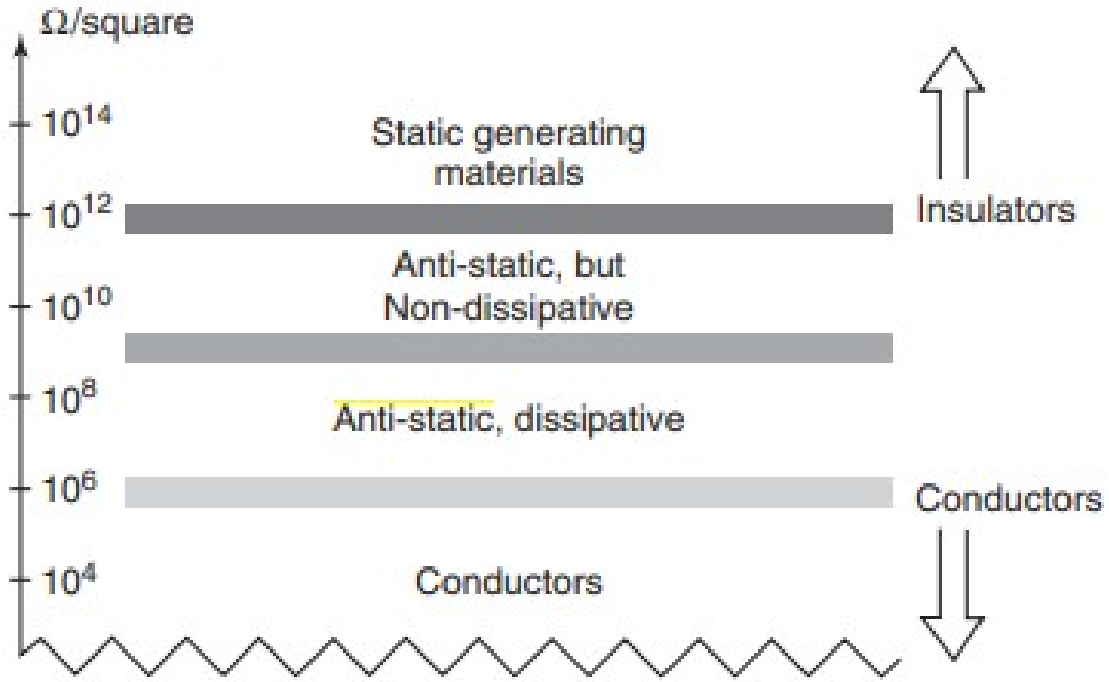



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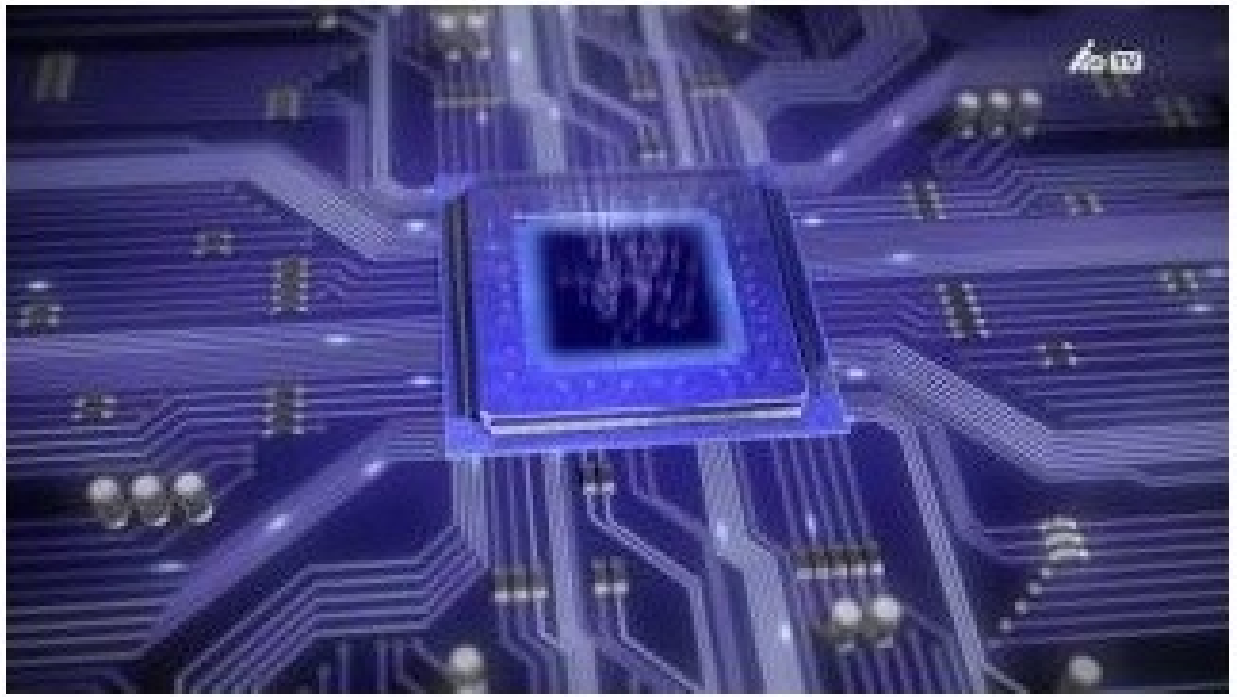



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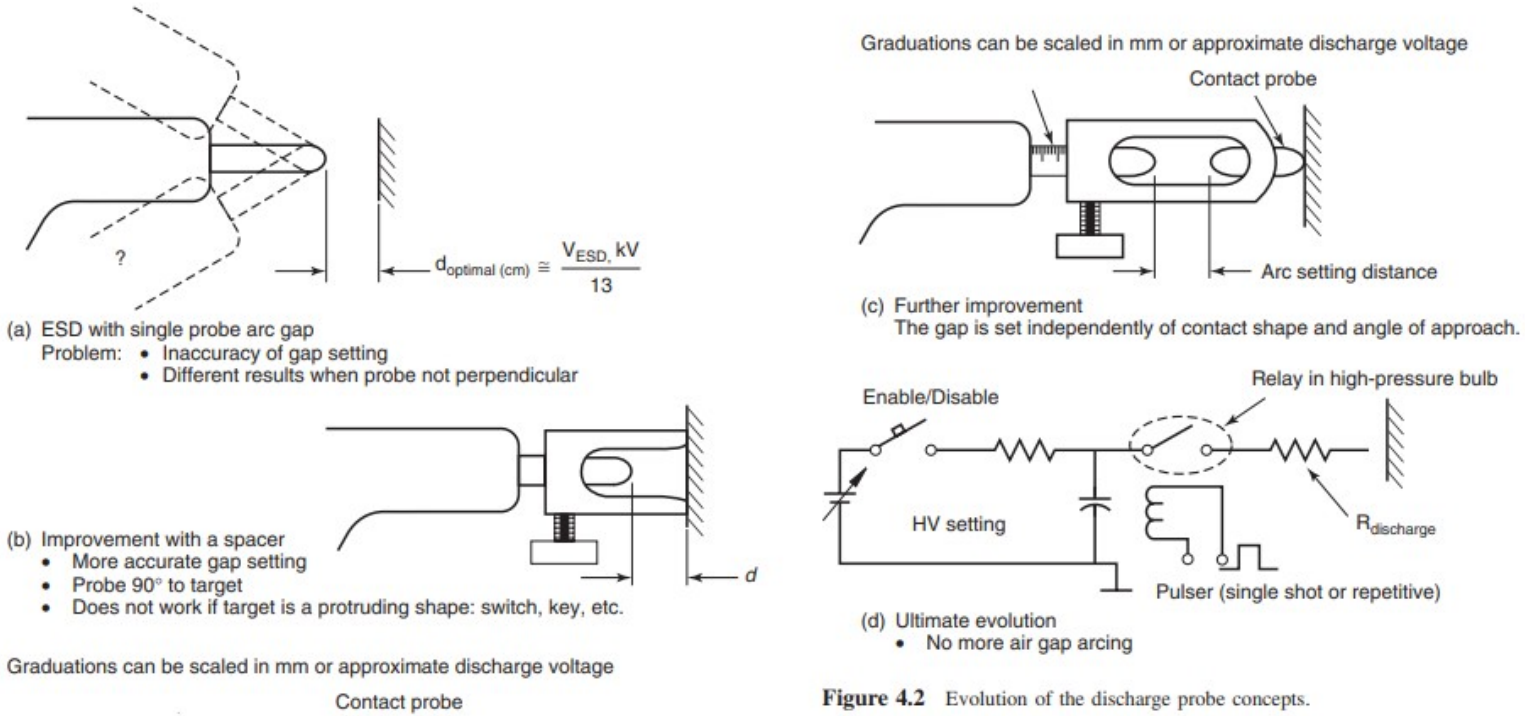



**Figure 4.2** Evolution of the discharge probe concepts.

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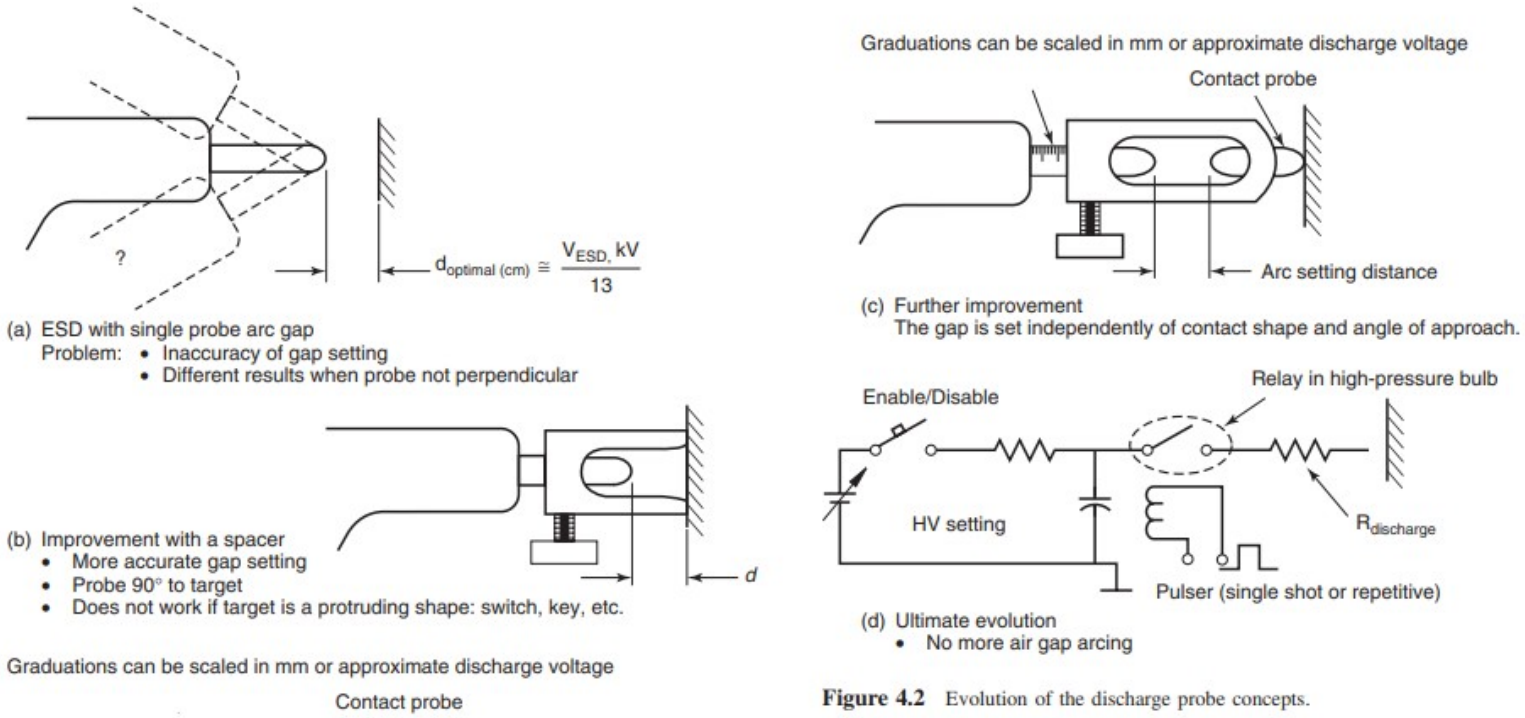



**Figure 4.2** Evolution of the discharge probe concepts.

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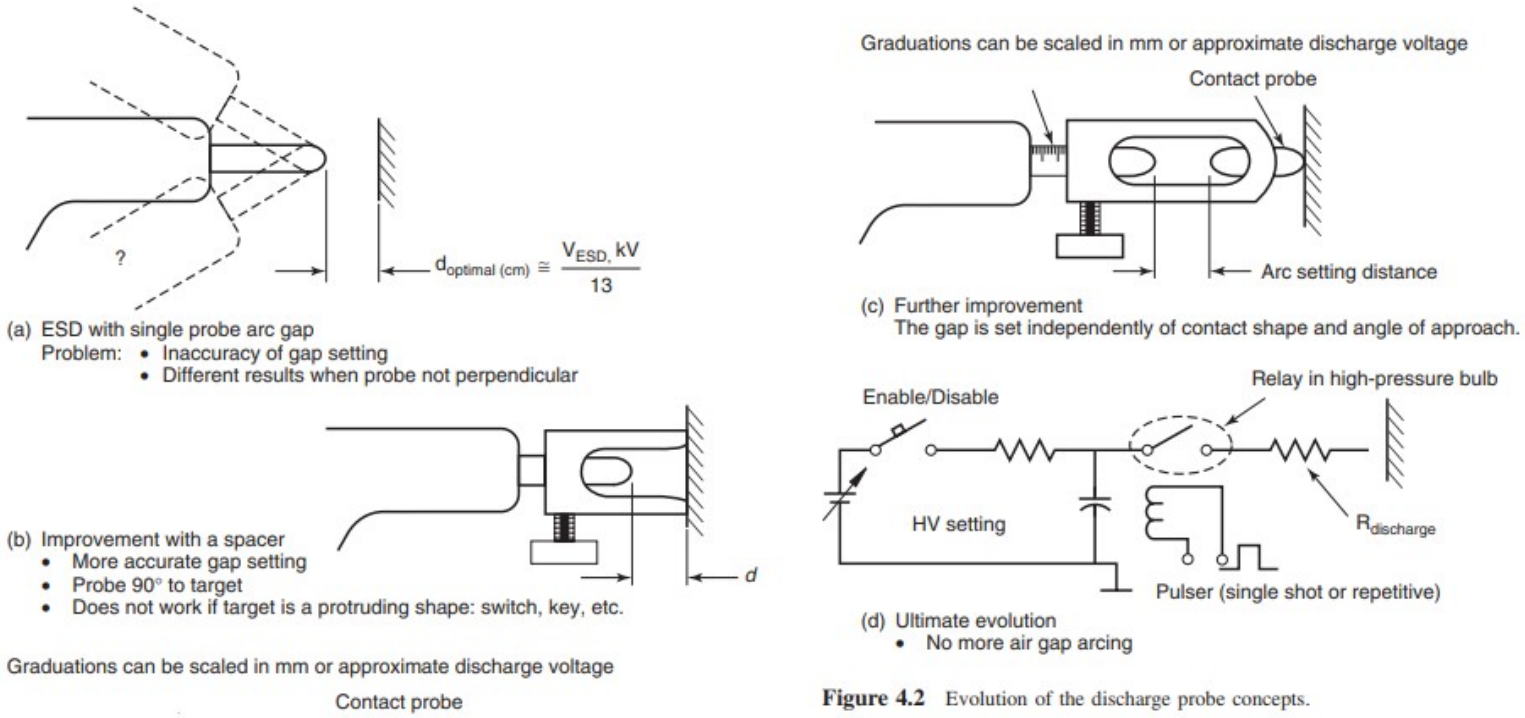



**Figure 4.2** Evolution of the discharge probe concepts.

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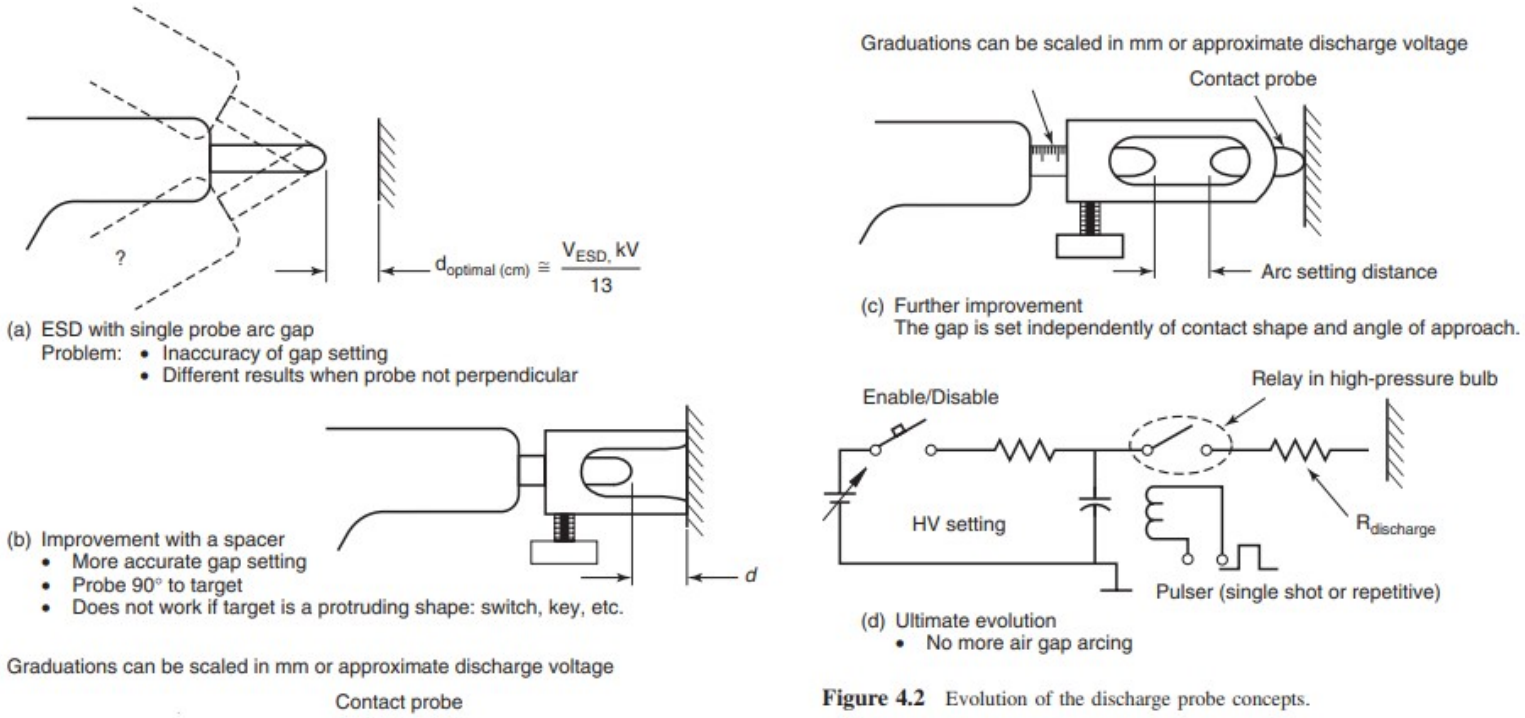



**Figure 4.2** Evolution of the discharge probe concepts.

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**Table 4.1** Arc Discharge vs. Contact Discharge, Pros and Cons[a]

| Topic | Arc ESD | Direct Contact ESD |
|---|---|---|
| Replicates real-life arc condition | +3 | |
| Guarantee of a well-reproducible waveform | | +3 |
| Cannot be performed on hard-to-reach recessed areas | | −3 |
| Requires scratching the paint | | −2 |
| Requires a carefully set gap dimension | −1 | |
| Insensitive to operator's body proximity and other human factors | | +2 |
| Insensitive to RH | | +1 |
| **Totals:** | **+2** | **+2** |

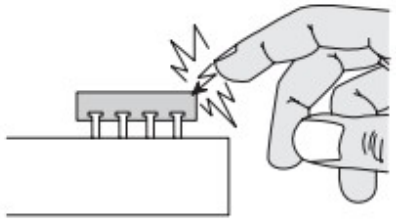
[a]Each topic is either a bonus or a penalty, but is never counted on both sides. A 1 (less important) to 3 (most important) weigthing factor has been used.

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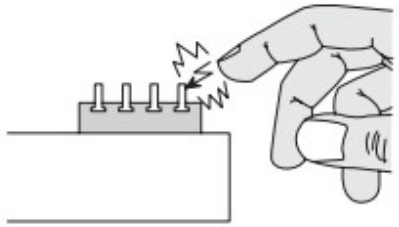
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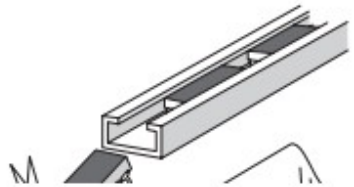
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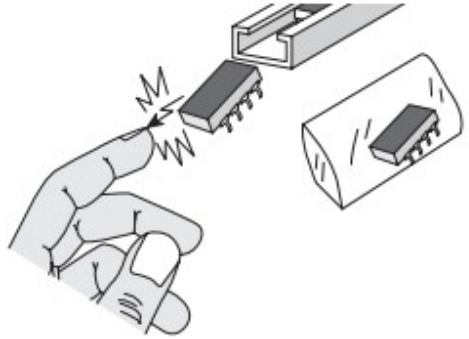
(a) Charged person zapping a module placed on a conductive workbench: discharge from finger → IC case → chip → pins



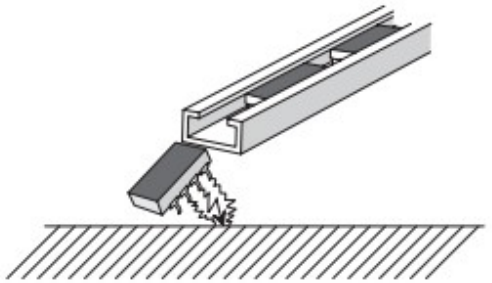
(b) Charged person zapping a module placed on a conductive workbench: discharge from finger → pin → chip → IC case.



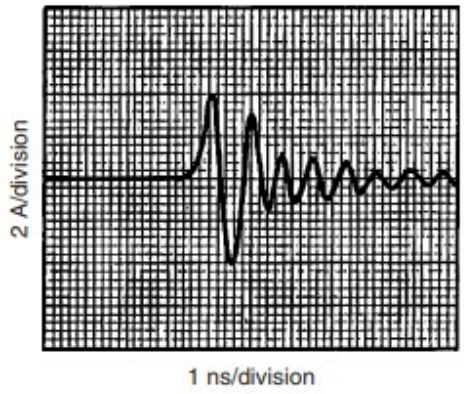
(c) Charged modules (from non-conductive plastic bags, rails, crates etc…) discharging on grounded personnel.



crates etc…) discharging on grounded personnel.



(d) Charged IC discharging on itself (charged device model)

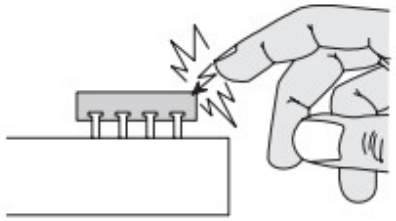



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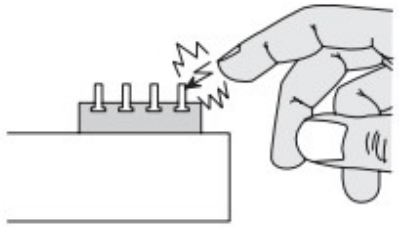
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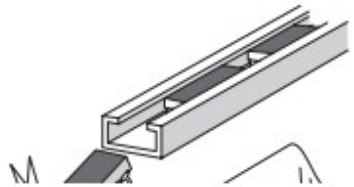
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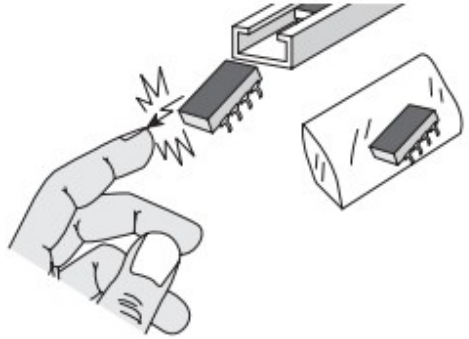
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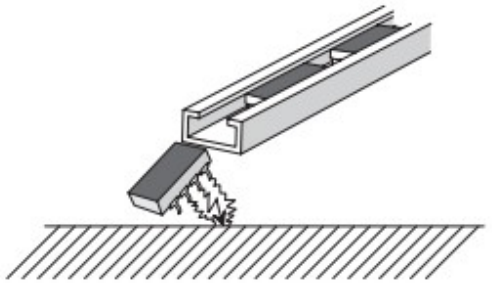
(b) Charged person zapping a module placed on a conductive workbench: discharge from finger → pin → chip → IC case.



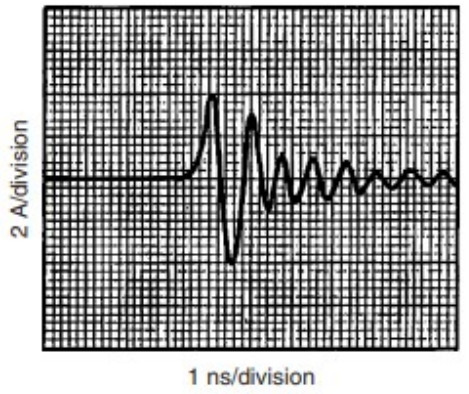
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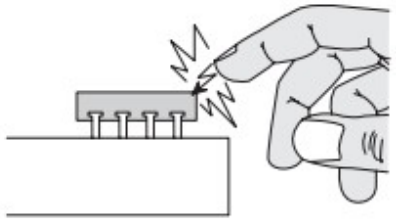



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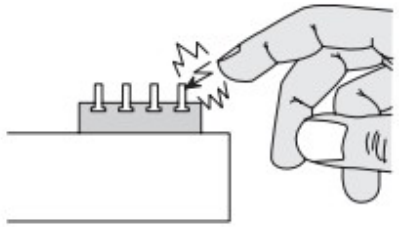
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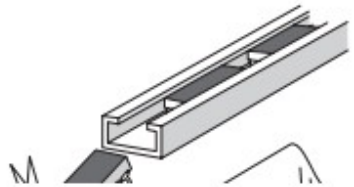
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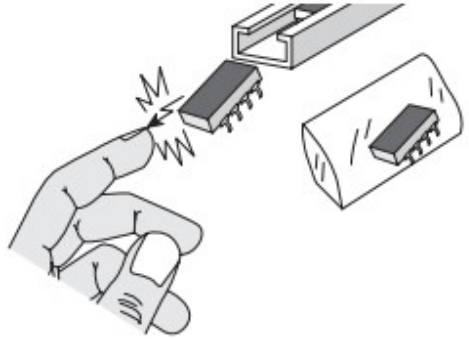
(a) Charged person zapping a module placed on a conductive workbench: discharge from finger → IC case → chip → pins



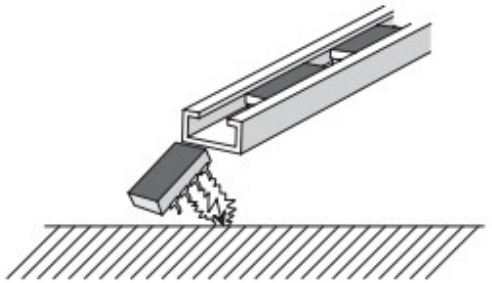
(b) Charged person zapping a module placed on a conductive workbench: discharge from finger → pin → chip → IC case.



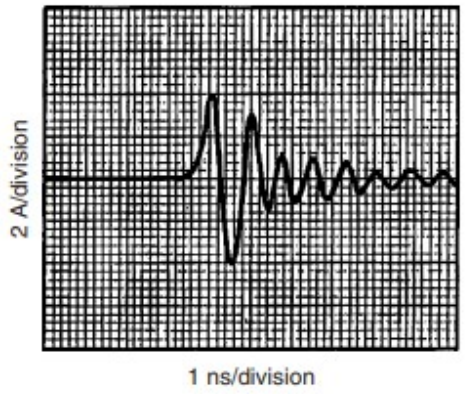
(c) Charged modules (from non-conductive plastic bags, rails, crates etc…) discharging on grounded personnel.



crates etc…) discharging on grounded personnel.



(d) Charged IC discharging on itself (charged device model)

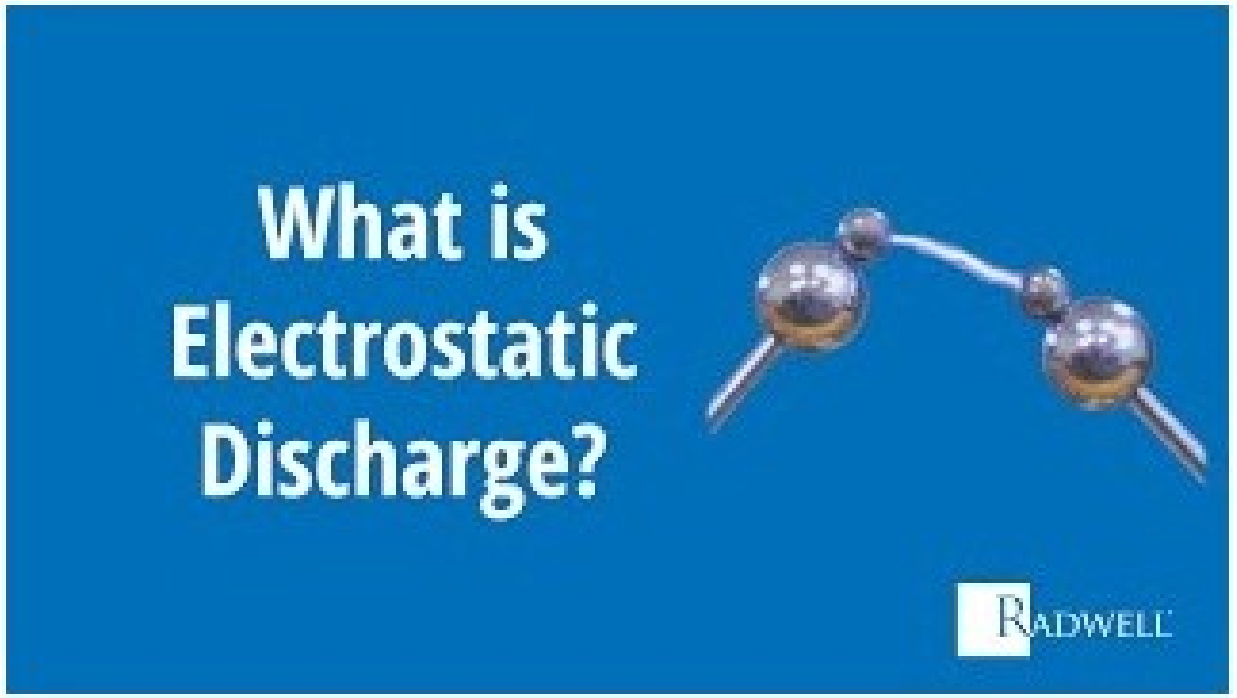



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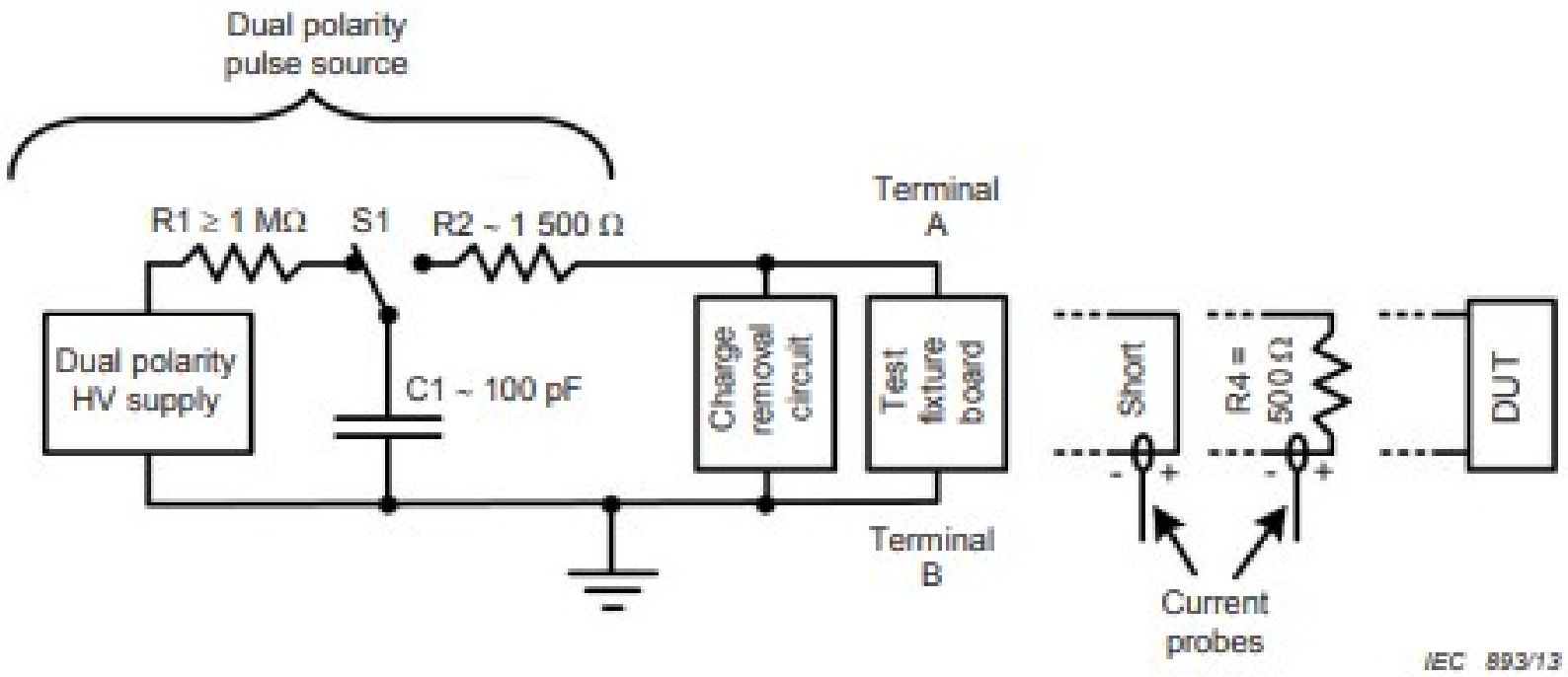



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**Table 4 – HBM ESD component classification levels**

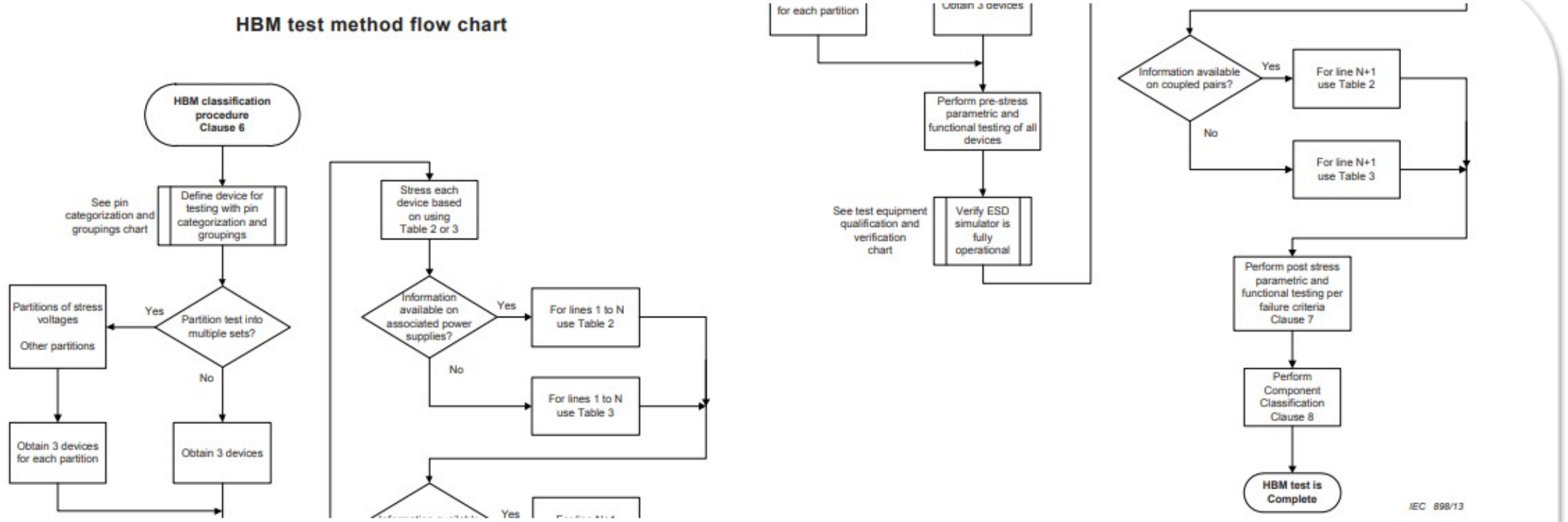
| Classification | Voltage Range (V) |
|---|---|
| 0A | < 125 |
| 0B | 125 to < 250 |
| 1A | 250 to < 500 |
| 1B | 500 to < 1 000 |
| 1C | 1 000 to < 2 000 |
| 2 | 2 000 to < 4 000 |
| 3A | 4 000 to < 8 000 |
| 3B | ≥ 8 000 |

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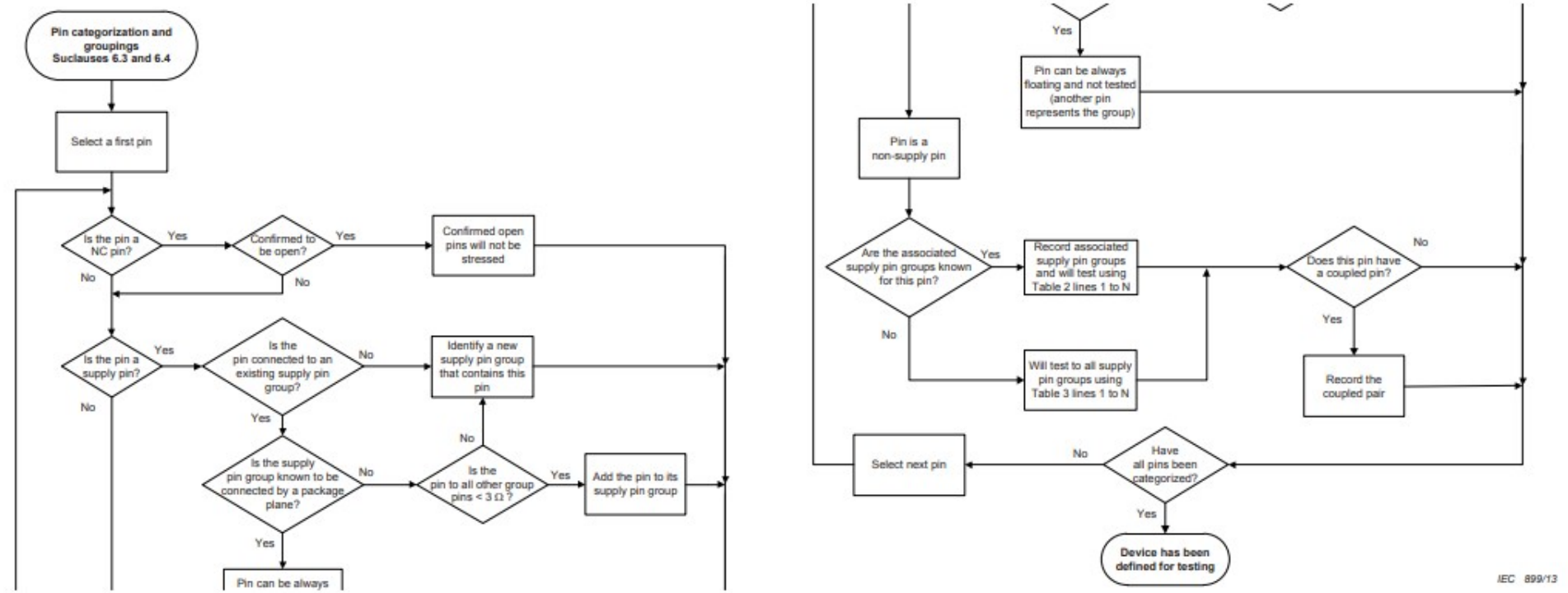



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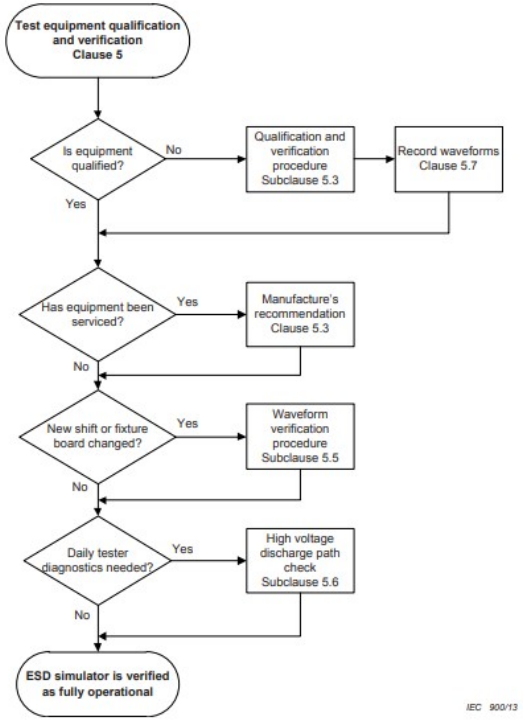



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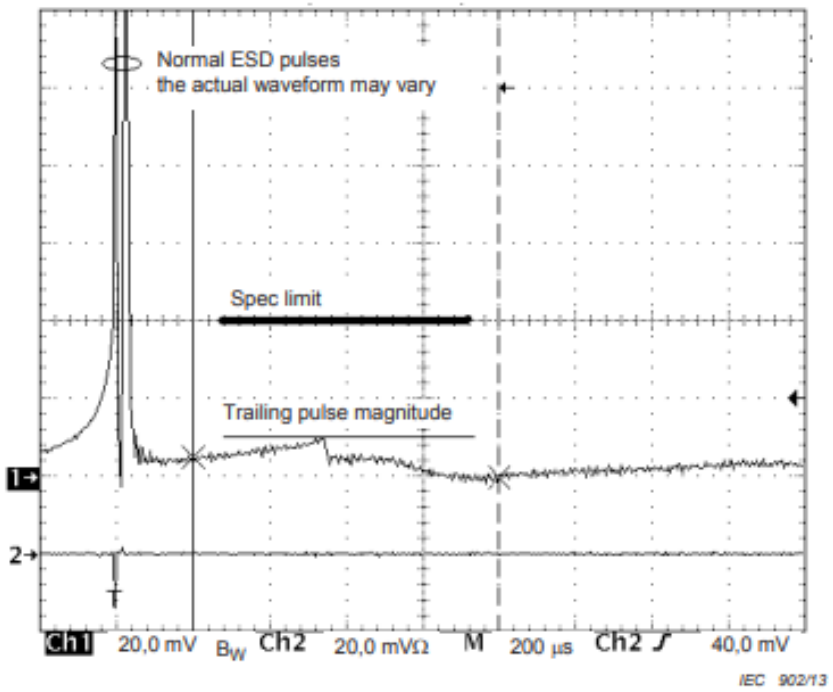



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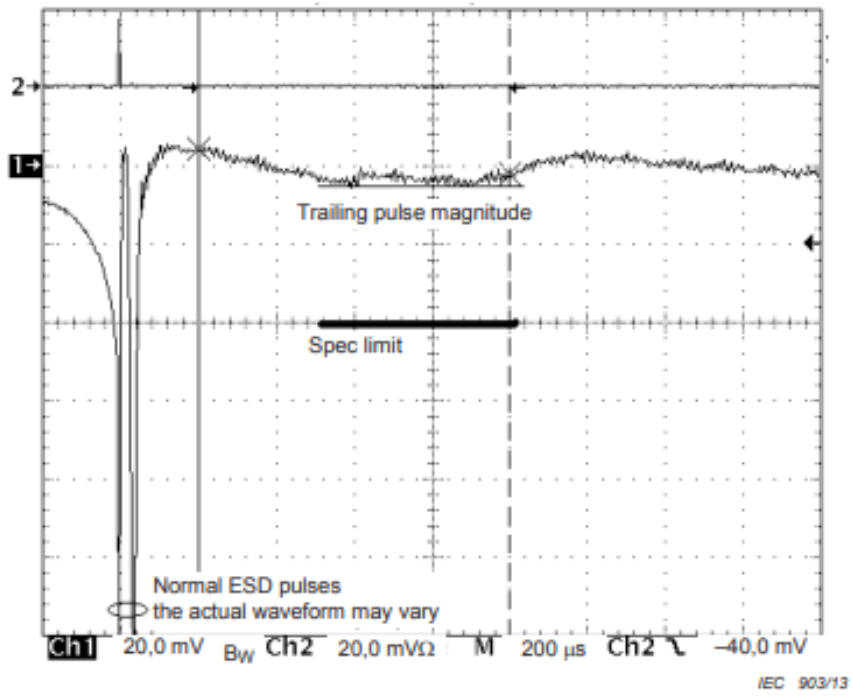



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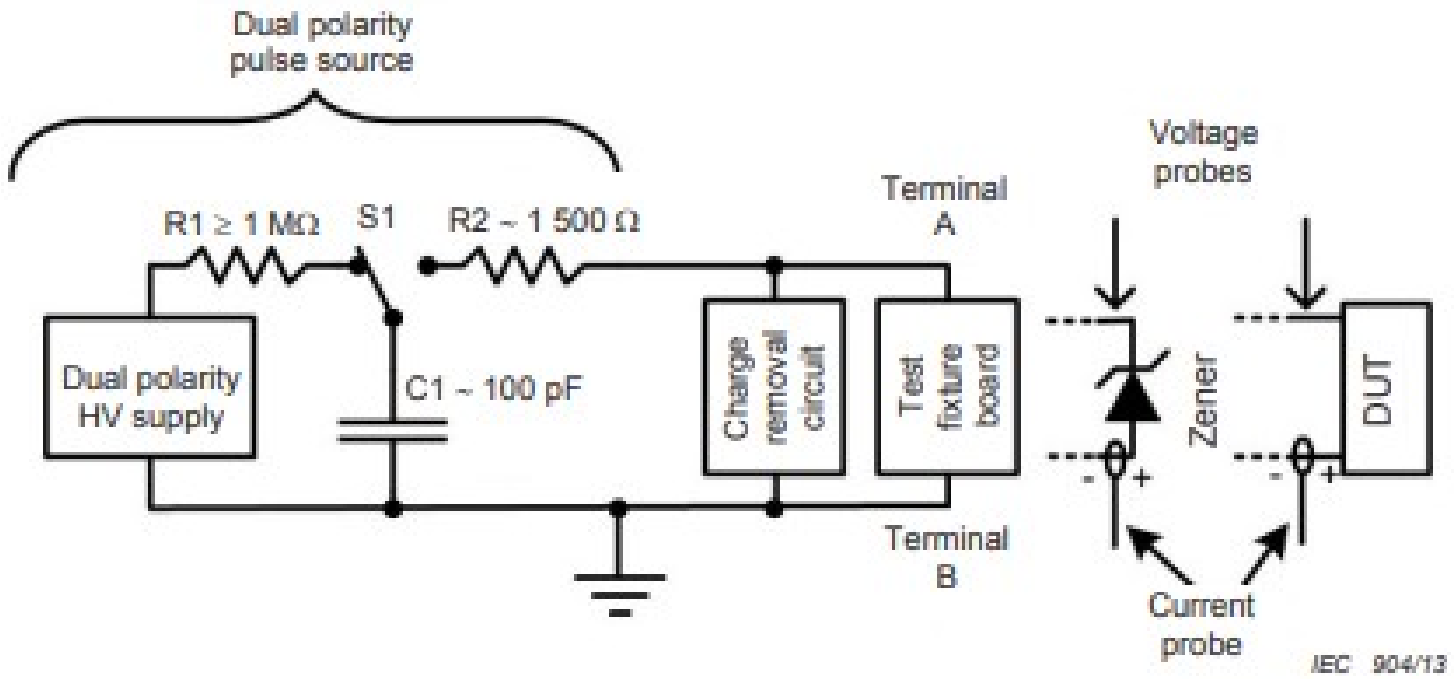



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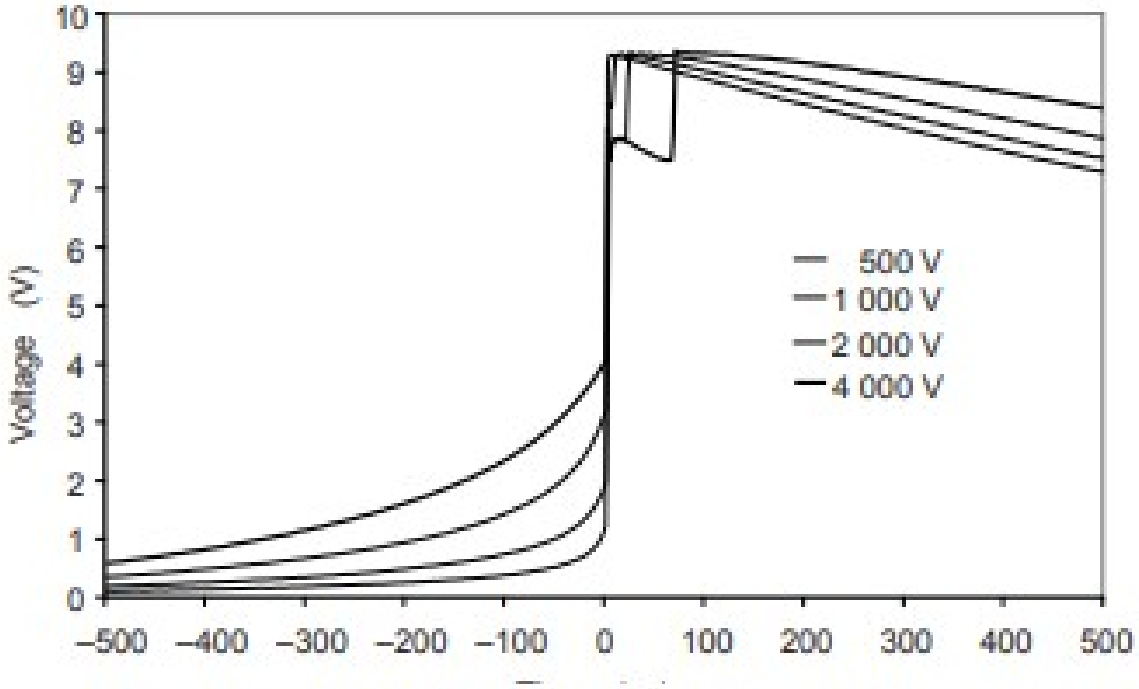



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**Table 3.1** Summary of Current and Proposed ESD Standards

| Category | Max. Voltage Pol. +/− | Discharge Network | Comments |
|---|---|---|---|
| For Devices | | | |
| MIL-Std 883 | 2 kV(Class 1) up to 8 kV | 100 pF/1500 Ω | Human Body Model (HBM) |
| DOD Std 1686 & Handbook 263 | N.A. | | ESD control program ESD protection guidelines |
| EIA/JEDEC Test Method 5.1 HBM | Up to 8 kV | 100 pF/1500 Ω | Used for device sensitivity evaluation (nonmandatory) |
| Test Method 5.2 MM | Up to 800 V | 15 Ω/0.5 μH/200 pF | (theoretically, $R = 0$) |
| Test Method 5.3 CDM | Up to 2 kV | 30 Ω/30 nH/10 pF | $R,L,C$ parameters may vary depending on IC package |
| IEC 61.340-5.1 and 5.2 replaces former 1000.15.1 | N.A | | Antistatic control procedures Approach similar to MIL-Std |
| For Equipment | | | |
| IEC 61000.4.2 (former IEC 801-2) | Up to 8 kV (contact) 15,000 V (Air) | 330 Ω/150 pF | Mandatory, part of EMC regulatory compliance |
| ANSI C.63-16 | Up to 8 kV (contact) 15,000 V (Air) | 330 Ω/150 pF | Similar to IEC, with differences in simulator and test procedure |
| ECMA 237 (from TC20) | | | Follows IEC 61000.4.2 |
| SAE J1113 Part13 & ISO 10.605 (Automobile) | 4–8 kV (contact) 4–25 kV (Air) | 2 kΩ/330 pF or 330 Ω/150 pF | Different test for modules and for vehicle install. |
| MIL-Std 1541 Spacecraft | 10 kV & more | Arc source | Network depends on application |
| STANAG 4235, 4239 (NATO Std) | 25 kV 300 kV | 500 Ω/500 pF 1 Ω/1 μH/1 nF | Personnel Helicopter discharge |

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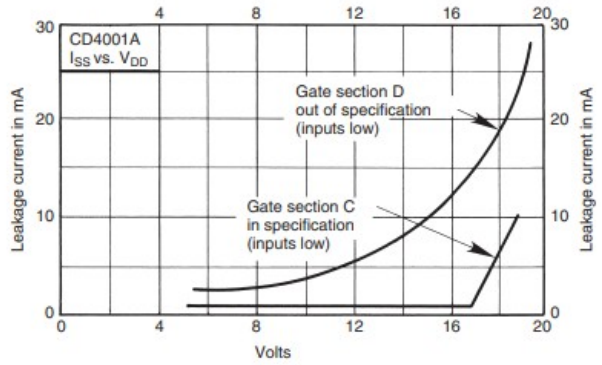



**Figure 2.5** Example of deteriorations in reverse breakdown (9).

**Table 2.2** Failure Decision Is Never a Step Function (11)

| Pulse Level (V) | Number of Pulses to Cause Failure ($N$) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 8 | 50 | 75 | 125 | 175 | 200 |
| 5750 | 10 | | | | | | | | | |
| 5000 | 12 | 1 | | | 2 | | | | | |
| 4250 | 16 | | | | 1 | | | | | 3 |
| 3500 | 9 | | | 2 | 1 | | | | | 8 |
| 2750 | 2 | | 1 | | 1 | 1 | 1 | 1 | 2 | 11 |
| 2000 | | | | | | | | | | 10 |

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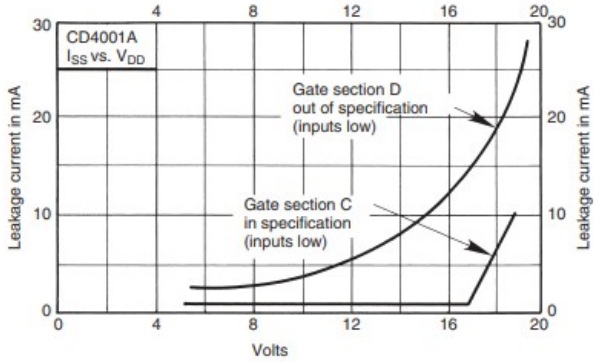



**Figure 2.5** Example of deteriorations in reverse breakdown (9).

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| Pulse Level (V) | Number of Pulses to Cause Failure ($N$) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 8 | 50 | 75 | 125 | 175 | 200 |
| 5750 | 10 | | | | | | | | | |
| 5000 | 12 | 1 | | | 2 | | | | | |
| 4250 | 16 | | | | 1 | | | | | 3 |
| 3500 | 9 | | | 2 | 1 | | | | | 8 |
| 2750 | 2 | | 1 | | 1 | 1 | 1 | 1 | 2 | 11 |
| 2000 | | | | | | | | | | 10 |

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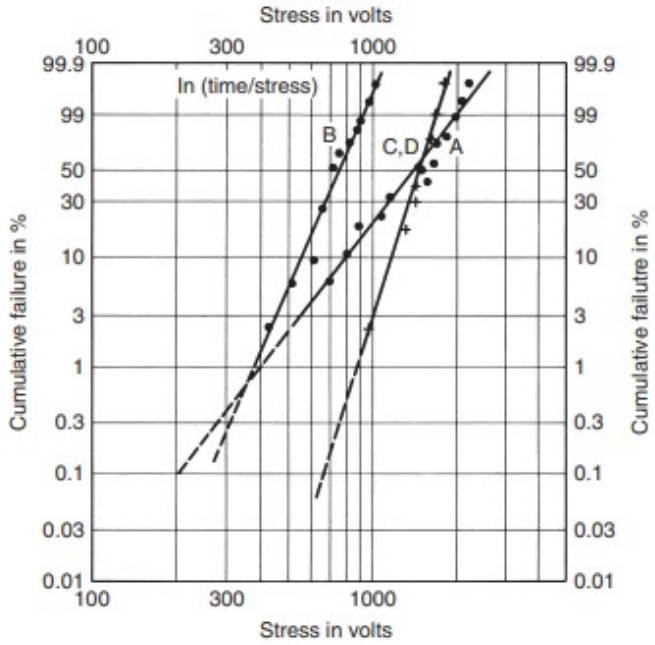



**Figure 2.6** ESD failure level for 4001 inverters from Manufacturers A, B, C, and D (10).

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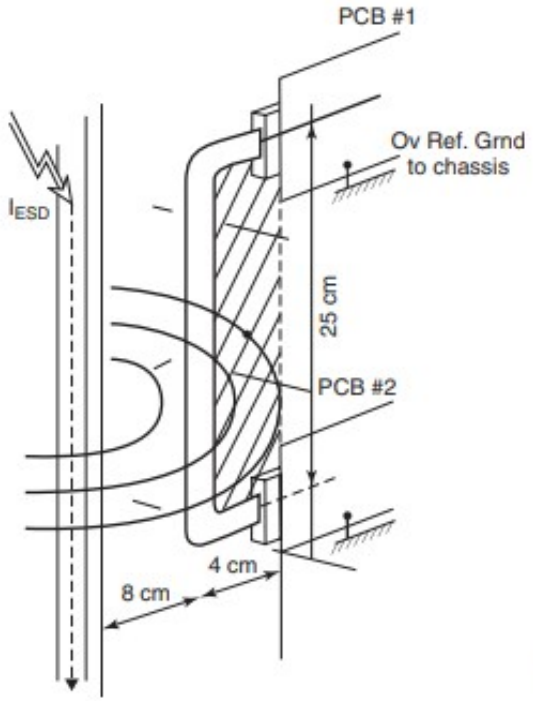


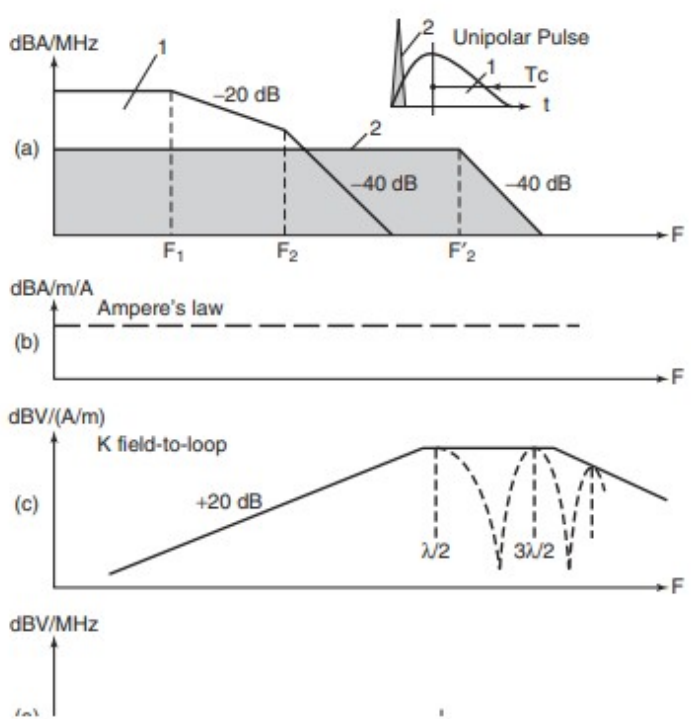

**Figure E.1** Coupling example of a D-ESD illuminating a nearby cable.

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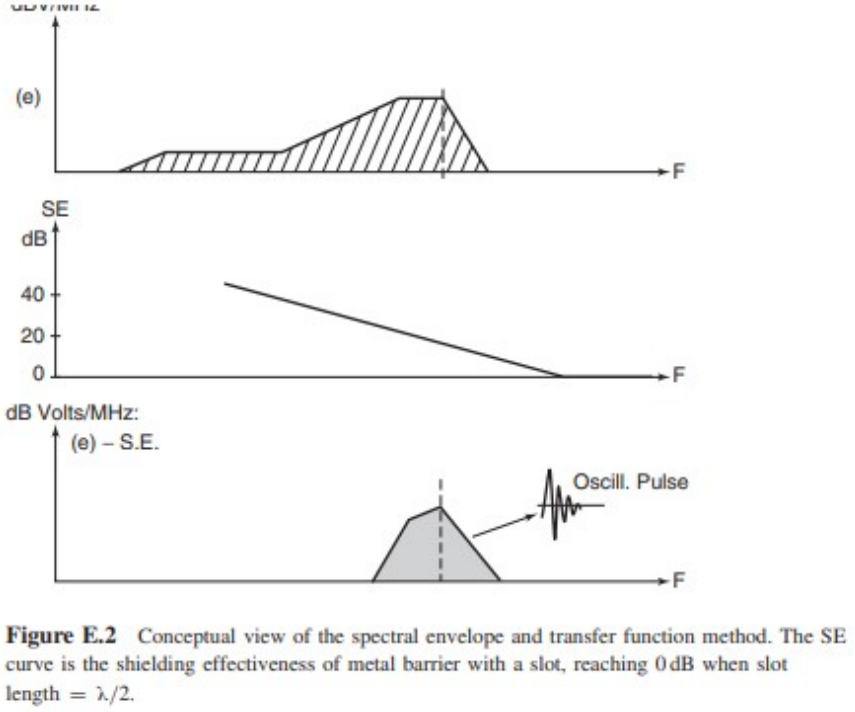




**Figure E.2** Conceptual view of the spectral envelope and transfer function method. The SE curve is the shielding effectiveness of metal barrier with a slot, reaching 0 dB when slot length = λ/2.

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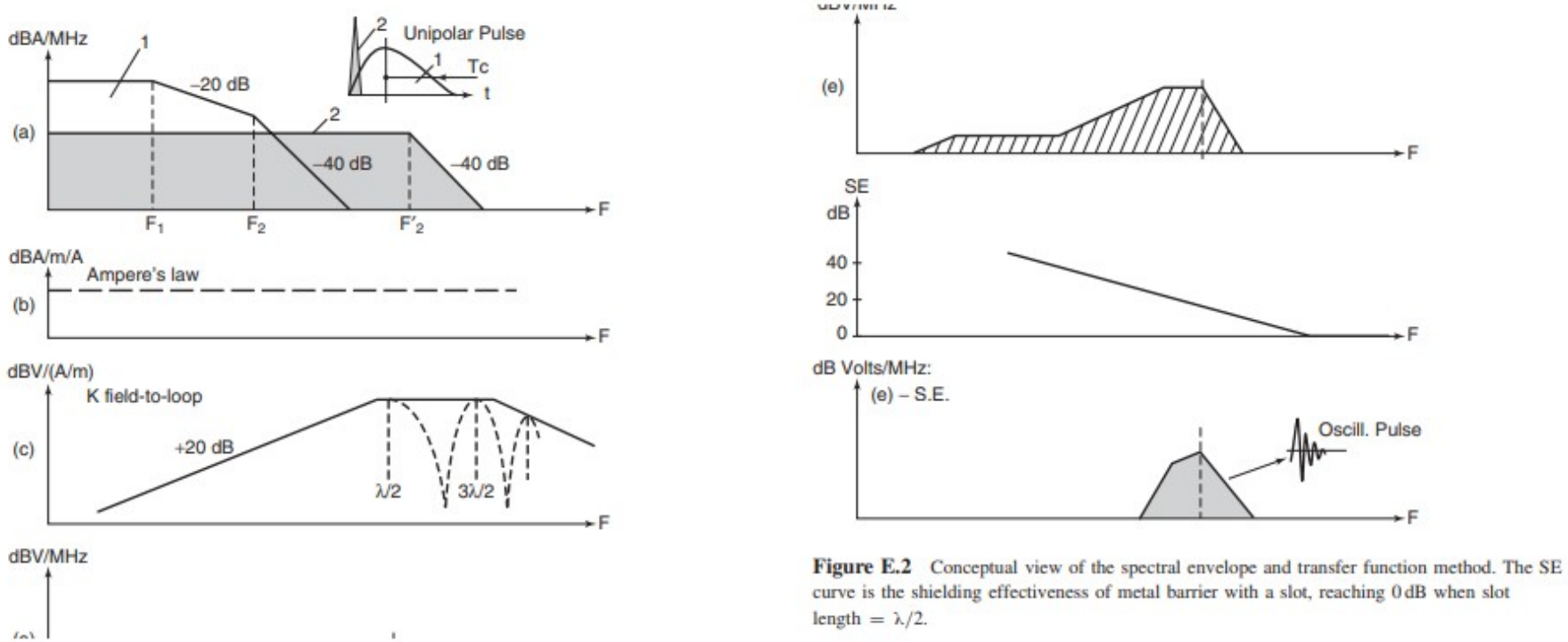



**Figure E.2** Conceptual view of the spectral envelope and transfer function method. The SE curve is the shielding effectiveness of metal barrier with a slot, reaching 0 dB when slot length = λ/2.

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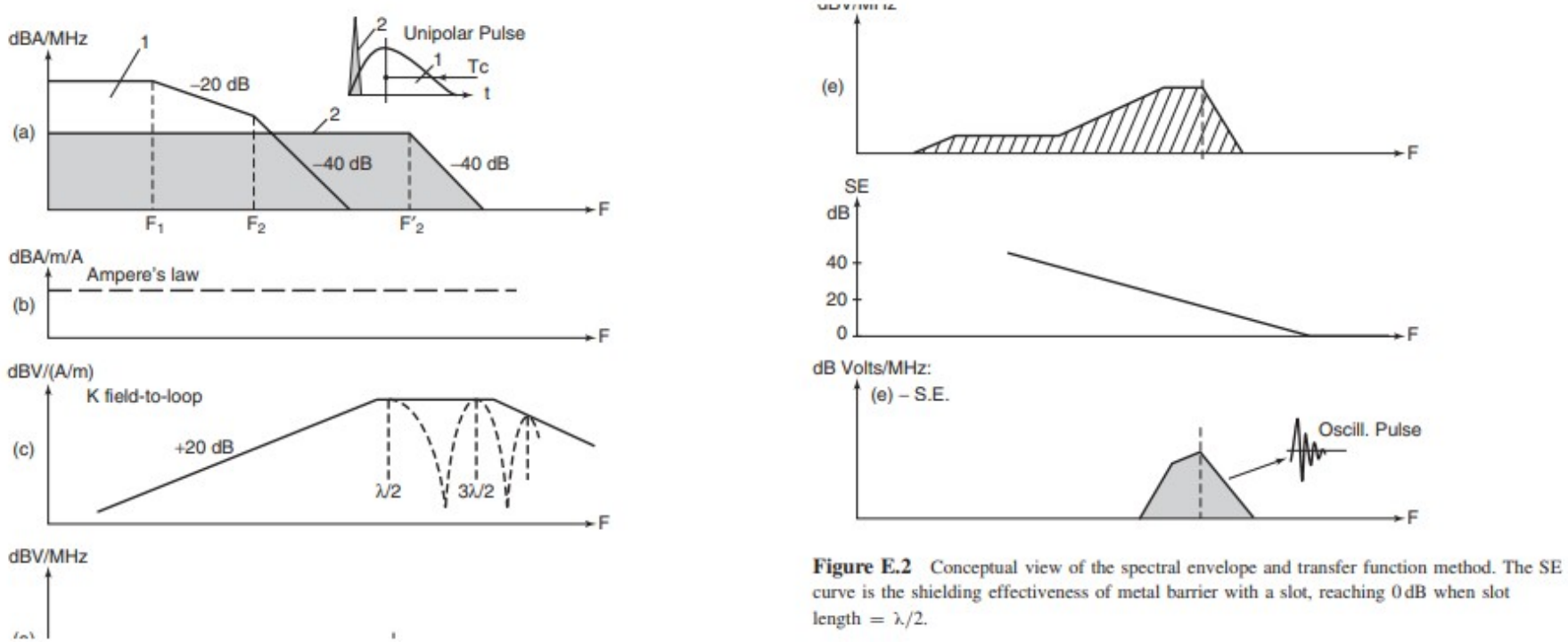



**Figure E.2** Conceptual view of the spectral envelope and transfer function method. The SE curve is the shielding effectiveness of metal barrier with a slot, reaching 0 dB when slot length = λ/2.

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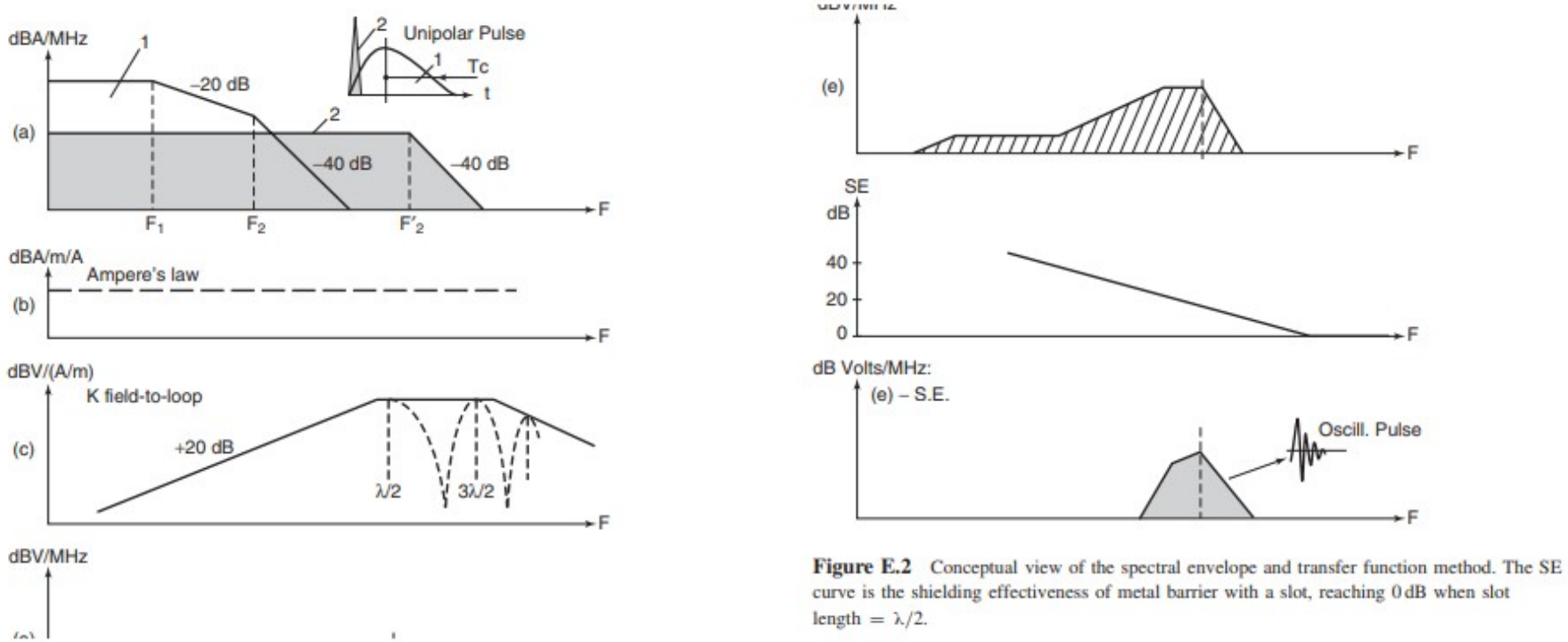



**Figure E.2** Conceptual view of the spectral envelope and transfer function method. The SE curve is the shielding effectiveness of metal barrier with a slot, reaching 0 dB when slot length = λ/2.

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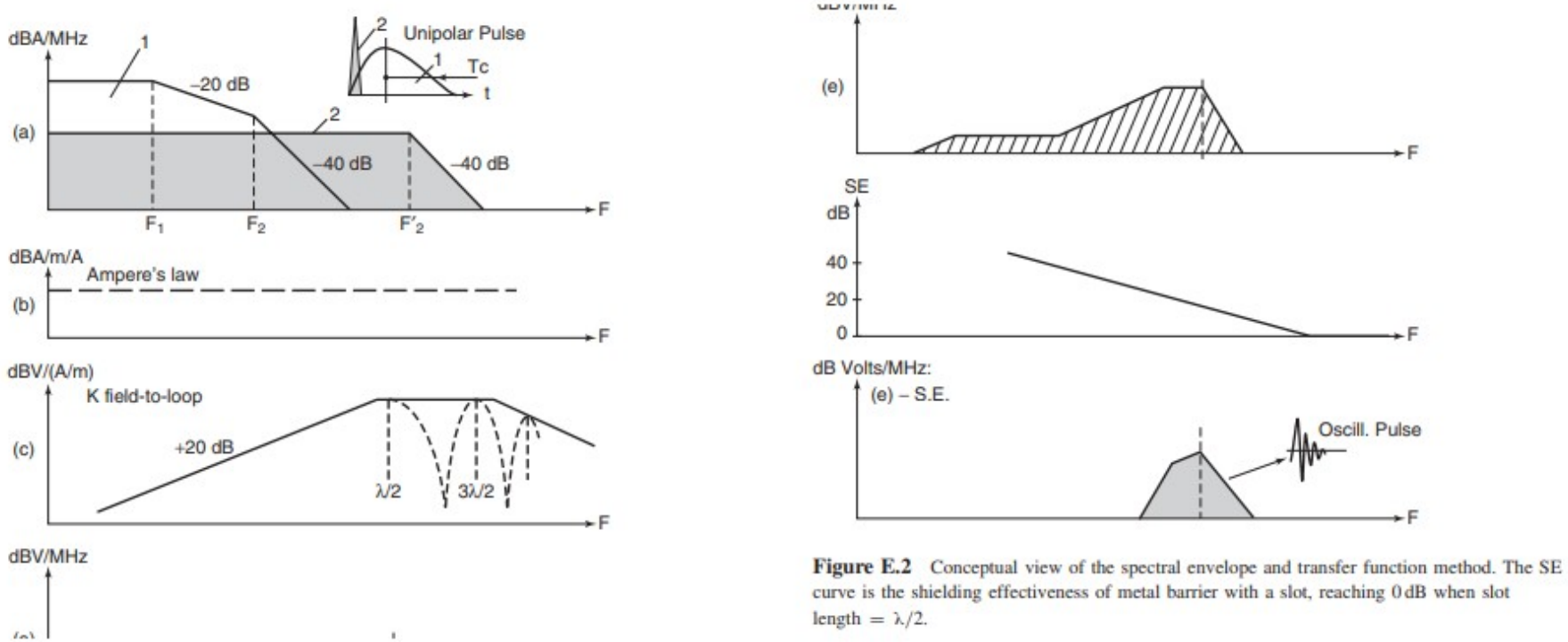



**Figure E.2** Conceptual view of the spectral envelope and transfer function method. The SE curve is the shielding effectiveness of metal barrier with a slot, reaching 0 dB when slot length = λ/2.

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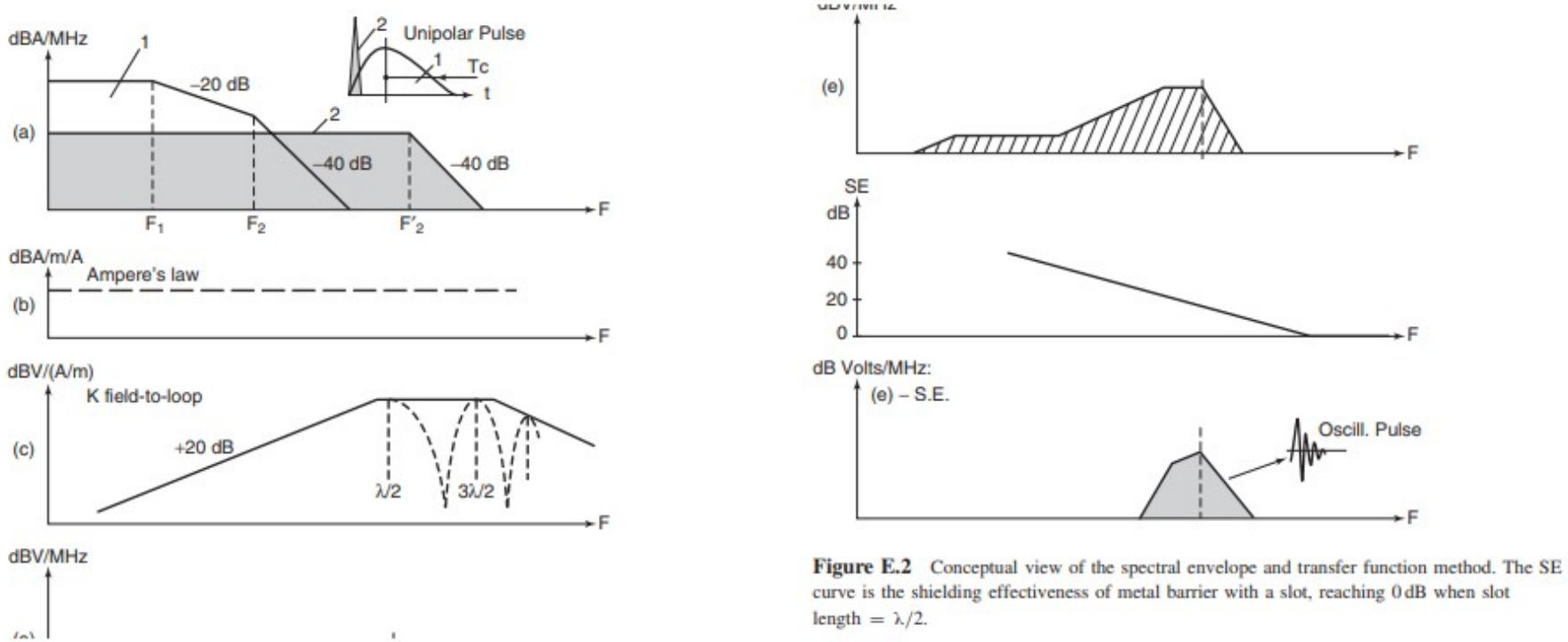


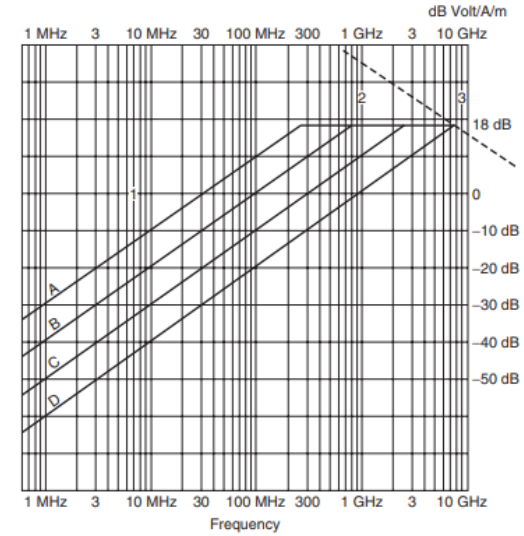

**Figure E.2** Conceptual view of the spectral envelope and transfer function method. The SE curve is the shielding effectiveness of metal barrier with a slot, reaching 0 dB when slot length = λ/2.

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**In region 1** ($l < \lambda/2$): K dBV/A/m = –62 + 20 log ($l$ cm × $h$ cm) + 20 logFMHz
**In region 2** (where $l > \lambda/2$): K = 18 dB + 20 log hcm
**In region 3** (where $l$ and $h$ are > $\lambda/2$): K = 95 – 20 log FMHz, for any value of "$h$"
**Curve A:** $l$ = 30 cm **Curve B:** $l$ = 10 cm **Curve C:** $l$ = 3 cm **Curve D:** $l$ = 1 cm
All curves made for h = 1 cm. For other values of $h$, add 20 log $h$(cm).

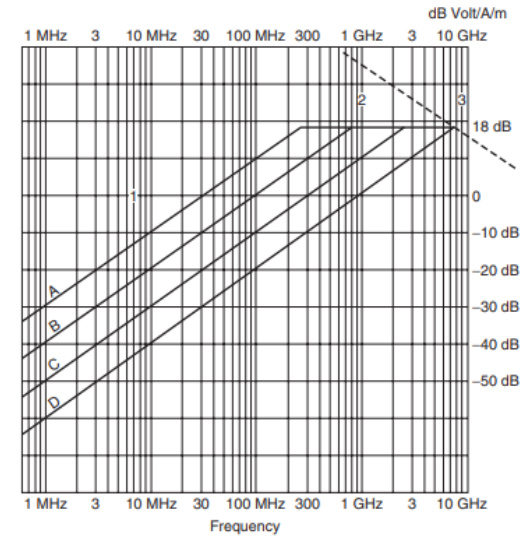
**Figure E.3** H-field-to-loop coupling vs. frequency, for uniform illumination of a loop with dimensions $l \times h$.

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**In region 1** ($l < \lambda/2$): K dBV/A/m = –62 + 20 log ($l$ cm × $h$ cm) + 20 logFMHz
**In region 2** (where $l > \lambda/2$): K = 18 dB + 20 log hcm
**In region 3** (where $l$ and $h$ are > $\lambda/2$): K = 95 – 20 log FMHz, for any value of "$h$"
**Curve A:** $l$ = 30 cm **Curve B:** $l$ = 10 cm **Curve C:** $l$ = 3 cm **Curve D:** $l$ = 1 cm
All curves made for h = 1 cm. For other values of $h$, add 20 log $h$(cm).

**Figure E.3** H-field-to-loop coupling vs. frequency, for uniform illumination of a loop with dimensions $l \times h$.

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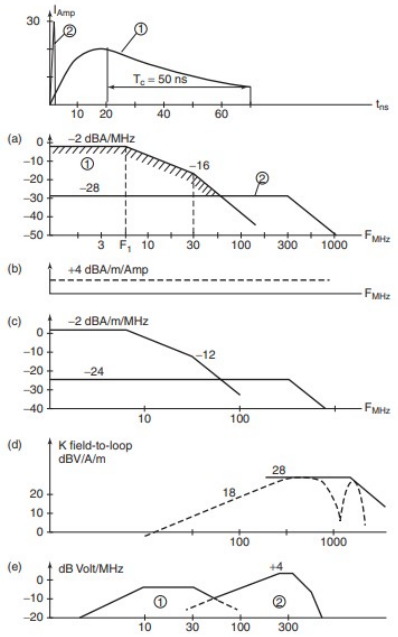



**Figure E.4** Detailed graphical process for the ESD-to-cable coupling example. (**a**) Spectral density of waveforms 1 and 2; (**b**) current-to-*H*-field conversion, for a 10.5-cm distance; (**c**) *H*-field spectrum density resulting of (a) + (b); (**d**) Field-to-loop coupling factor for a 25 × 4 cm loop; (**e**) final voltage spectrum, resulting of (c) + (d).

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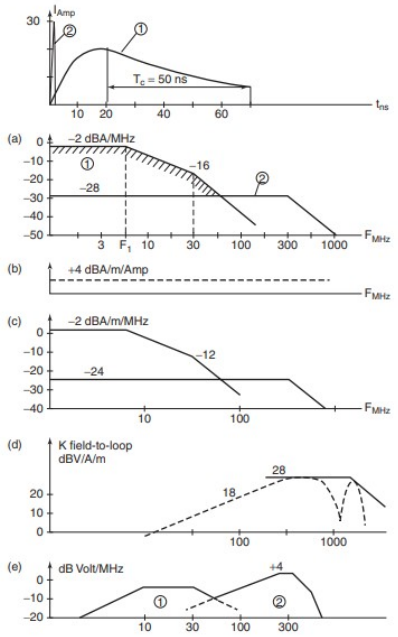



**Figure E.4** Detailed graphical process for the ESD-to-cable coupling example. (**a**) Spectral density of waveforms 1 and 2; (**b**) current-to-*H*-field conversion, for a 10.5-cm distance; (**c**) *H*-field spectrum density resulting of (a) + (b); (**d**) Field-to-loop coupling factor for a 25 × 4 cm loop; (**e**) final voltage spectrum, resulting of (c) + (d).

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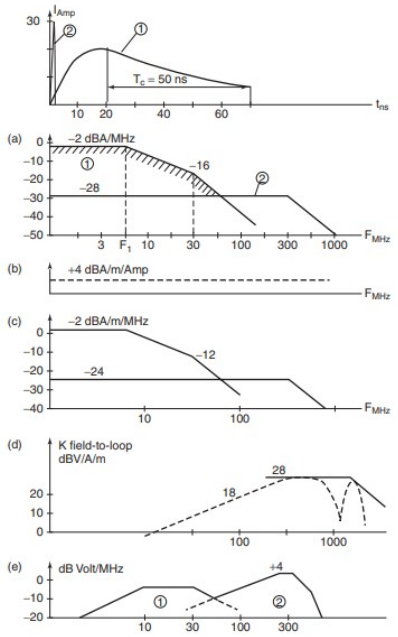



**Figure E.4** Detailed graphical process for the ESD-to-cable coupling example. (**a**) Spectral density of waveforms 1 and 2; (**b**) current-to-*H*-field conversion, for a 10.5-cm distance; (**c**) *H*-field spectrum density resulting of (a) + (b); (**d**) Field-to-loop coupling factor for a 25 × 4 cm loop; (**e**) final voltage spectrum, resulting of (c) + (d).

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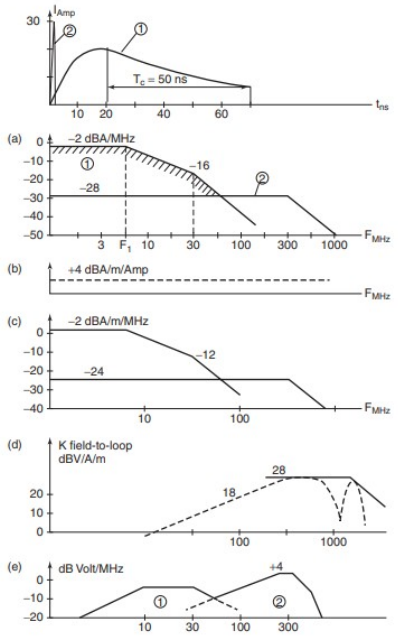



**Figure E.4** Detailed graphical process for the ESD-to-cable coupling example. (**a**) Spectral density of waveforms 1 and 2; (**b**) current-to-*H*-field conversion, for a 10.5-cm distance; (**c**) *H*-field spectrum density resulting of (a) + (b); (**d**) Field-to-loop coupling factor for a 25 × 4 cm loop; (**e**) final voltage spectrum, resulting of (c) + (d).

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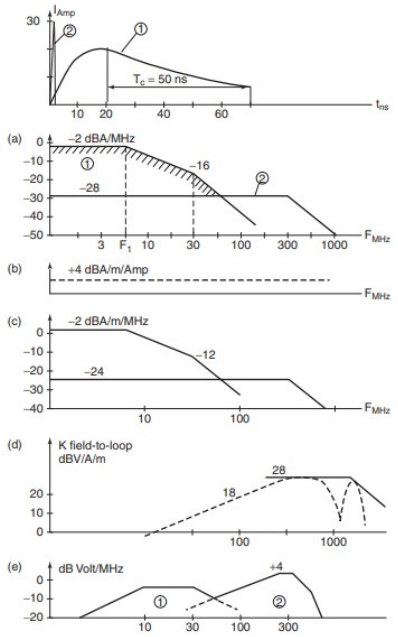



**Figure E.4** Detailed graphical process for the ESD-to-cable coupling example. (**a**) Spectral density of waveforms 1 and 2; (**b**) current-to-*H*-field conversion, for a 10.5-cm distance; (**c**) *H*-field spectrum density resulting of (a) + (b); (**d**) Field-to-loop coupling factor for a 25 × 4 cm loop; (**e**) final voltage spectrum, resulting of (c) + (d).

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**Table E.1** Detailed Calculation Steps of Frequency-Domain Coupling Prediction

| | Field-to-Cable Broadband Prediction by Frequency | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | 0.1 MHz | 1 MHz | 6 MHz | 10 MHz | 30 MHz | 100 MHz | 300 MHz | 550 MHz | 1000 MHz |
| **(A)** Spectral Density (dB A/MHz) | | | | | | | | | |
| Long pulse (1) | **−2** | **−2** | **−2** | **−6** | **−16** | −36 | −56 | | |
| Short pulse (2) | −28 | −28 | −28 | −28 | −28 | **−28** | **−28** | **−36** | **−48** |
| **(B)** H-field from 1 A wire @ 0.1 m distance (dB A/m/A) | +4 | +4 | +4 | +4 | +4 | +4 | +4 | +4 | +4 |
| **(D)** K field-to-loop (dBV/A/m) | −42 | −22 | −6 | −2 | +8 | +18 | +28 | +26 | +14 |
| **(E)** Final induced voltage density (dBV/MHz) = A+B+D | −40 | −20 | −4 | −4 | −4 | −6 | +4 | −6 | −26 |
| Density @ midband for partial integration | −30 | −12 | −4 | −4 | −11 | −1 | −6.5 | | |
| **(F)** Integral in frequency interval: Volts x $\Delta F$(MHz) | 0.03 V | 1.2 V | 3 V | 12 V | 20 V | 180 V | 325 V | | |

$\sum$ **of partial voltages integration, 0 → 1000 MHz: 540 V.**

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**Table E.2** Attenuation of Shield with a 10-cm Slot, Applied to Flat Cable Example

| Frequency Interval (MHz) | Voltage-Induced without Shield (V) | Average Attenuation of a 10-cm Slot (dB) | New Induced Voltage (V) | Global Attenuation |
|---|---|---|---|---|
| 1–10 | 4.2 | ⩾ 50 (or 300 times) | 0.014 | |
| 10–30 | 12 | ⩾ 35 (or 55 times) | 0.2 | |
| 30–100 | 20 | ⩾ 25 (18 times) | 10V | |
| 100–300 | 180 | ⩾ 15 (5.5 times) | 32 | |
| 300–1000 | 325 | ⩾ 5 (1.8 times) | 180 | |
| Total ≈ | 540 | | 220 | 2.5 (or 8 dB) |

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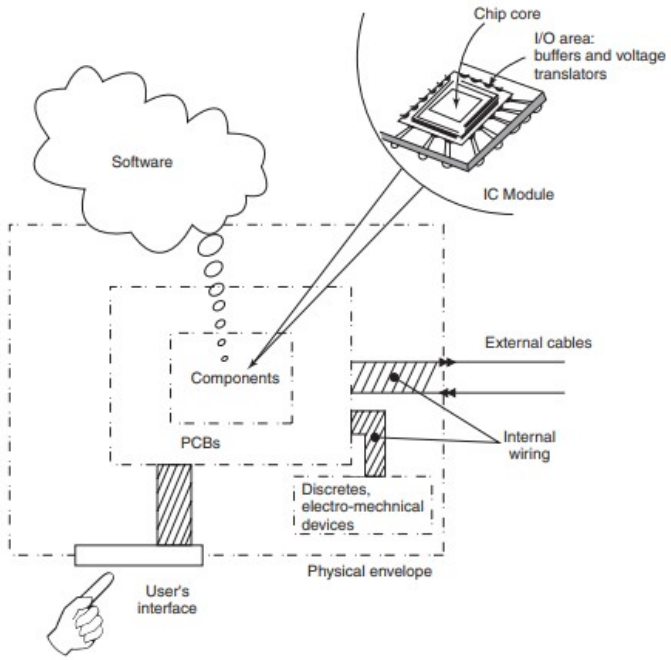


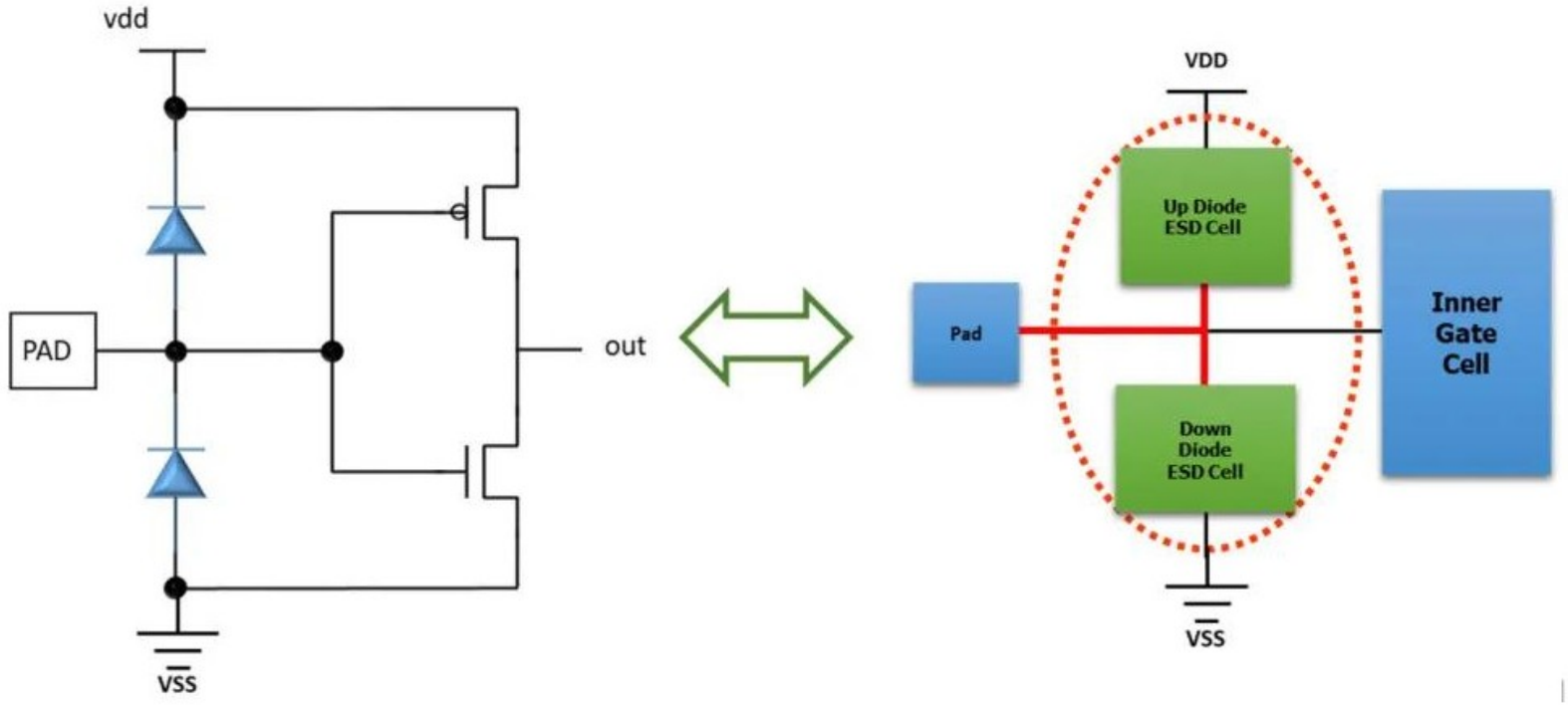

**Figure 5.1** Several layers to be considered for an ESD protection strategy.

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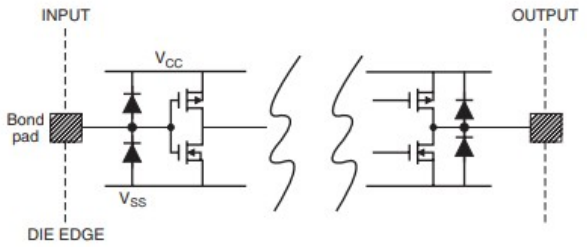



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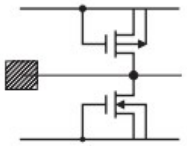
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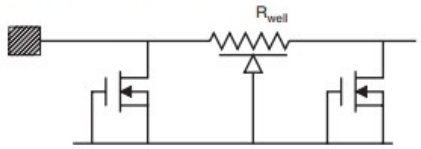
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(a) Simple Diodes Protection



(b) CMOS Transistor (Crowbar) Protection




(c) Resistor and Transistor Protection

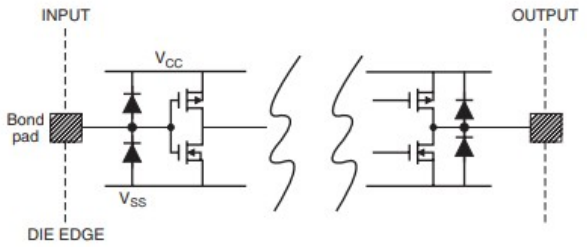
**Figure 5.2** Few widely used ESD protection schemes, embedded in the IC itself.

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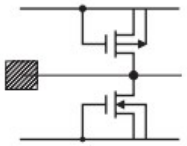
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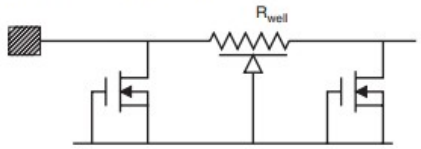
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(a) Simple Diodes Protection



(b) CMOS Transistor (Crowbar) Protection




(c) Resistor and Transistor Protection

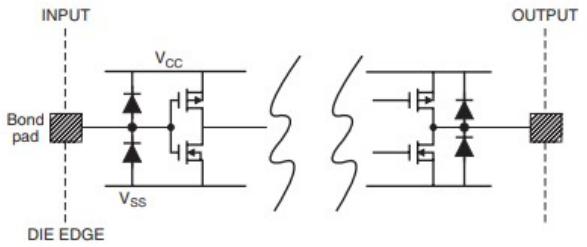
**Figure 5.2** Few widely used ESD protection schemes, embedded in the IC itself.

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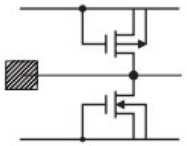
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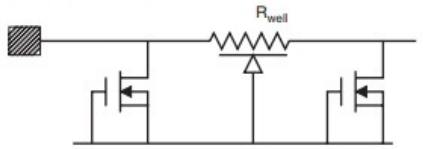
# Imported legacy visual reference




(a) Simple Diodes Protection



(b) CMOS Transistor (Crowbar) Protection




(c) Resistor and Transistor Protection

**Figure 5.2** Few widely used ESD protection schemes, embedded in the IC itself.

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**Table 5.1** Summary of Chip-Integrated ESD Protections

| IC Technology | Type of Embedded Protection | ESD Withstanding, No-Damage Level (HBM test, 150-ns pulse) |
|---|---|---|
| CMOS | Unprotected (bare chip) | 50 V (for 0.50-µm rule) |
| CMOS | Clamp diode + series resistance | 2–4 kV |
| Bipolar | Input protected by | |
| | Schottky diode (Vbr: 30 V) | 700 V |
| | Crowbar transistor | 2 kV |
| | Thyristor | 5 kV |
| | Output, unprotected | 2 kV |
| Low-speed drivers (RS 232) | Diodes + resistance + low-pass filter | Up to 15 kV |

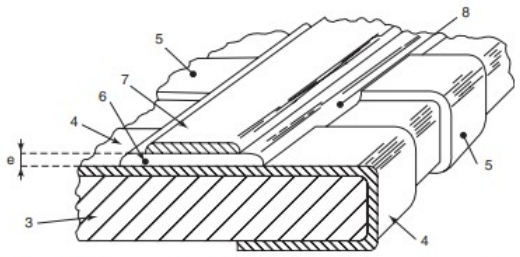



**Figure 5.3** Varistor "ring" for collective clamping of all the IC pins (6). The printed varistor ink trace (6) is crossing the chip leads (4, 5). On top of it, a grounding ring (7) collects the surge currents.

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**Table 5.1** Summary of Chip-Integrated ESD Protections

| IC Technology | Type of Embedded Protection | ESD Withstanding, No-Damage Level (HBM test, 150-ns pulse) |
|---|---|---|
| CMOS | Unprotected (bare chip) | 50 V (for 0.50-µm rule) |
| CMOS | Clamp diode + series resistance | 2–4 kV |
| Bipolar | Input protected by | |
| | Schottky diode (Vbr: 30 V) | 700 V |
| | Crowbar transistor | 2 kV |
| | Thyristor | 5 kV |
| | Output, unprotected | 2 kV |
| Low-speed drivers (RS 232) | Diodes + resistance + low-pass filter | Up to 15 kV |

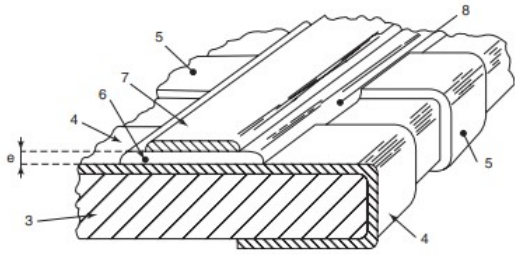



**Figure 5.3** Varistor "ring" for collective clamping of all the IC pins (6). The printed varistor ink trace (6) is crossing the chip leads (4, 5). On top of it, a grounding ring (7) collects the surge currents.

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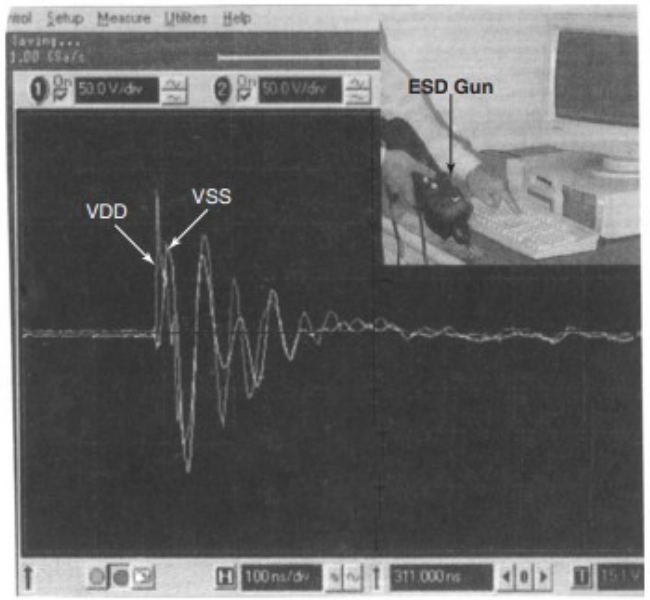


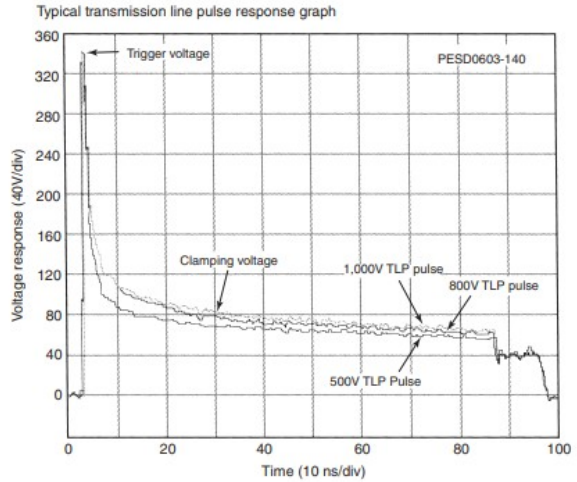

**Figure 5.4** Voltage at the $V_{cc}$ pin of a keyboard IC when a 1-kV indirect ESD is applied to the HCP at 10-cm distance. Although not destructive, the 130-V pulse causes a lockup of the keyboard in a frozzen state, requiring a power off/on action (7).

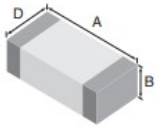
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Dimensions in millimeters *(inches)**

| | Length A | | Height B | |
|---|---|---|---|---|
| Part | Min. | Max. | Min. | Max. |
| **0402** | 0.95 *(0.037)* | 1.05 *(0.041)* | 0.33 *(0.010)* | 0.43 *(0.017)* |
| **0603** | 1.40 *(0.055)* | 1.80 *(0.071)* | 0.38 *(0.015)* | 0.58 *(0.023)* |

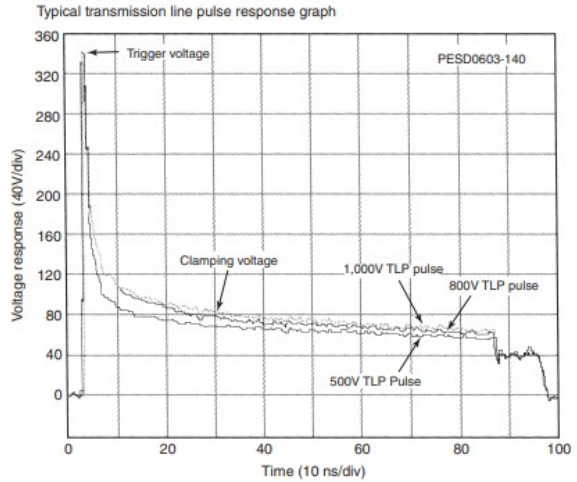
**Figure 5.5** Transient voltage suppressors, with very low parasitic capacitance. (Source Tyco/Raychem.)

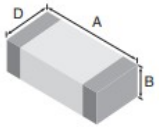
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Dimensions in millimeters *(inches)**

| | Length A | | Height B | |
|---|---|---|---|---|
| **Part** | **Min.** | **Max.** | **Min.** | **Max.** |
| **0402** | 0.95 *(0.037)* | 1.05 *(0.041)* | 0.33 *(0.010)* | 0.43 *(0.017)* |
| **0603** | 1.40 *(0.055)* | 1.80 *(0.071)* | 0.38 *(0.015)* | 0.58 *(0.023)* |

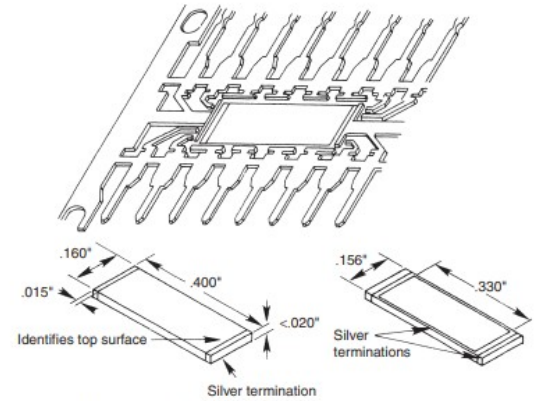
**Figure 5.5** Transient voltage suppressors, with very low parasitic capacitance. (Source Tyco/Raychem.)

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| | |
|---|---|
| Capacitance pin 8 to pin 16 | 0.1 to 0.5 $\mu$fd |
| Normal voltage | –10 V to + 10 V |
| Operating temperature | –55° C to + 125° C |
| Capacitance substrate to pin 16 (optional) | 0.001 to .005 $\mu$fd |
| Typ $\Delta V_{cc}$ for current transient of +20 mA/ns $V_{cc}$ to ground | +0.040 Volts |

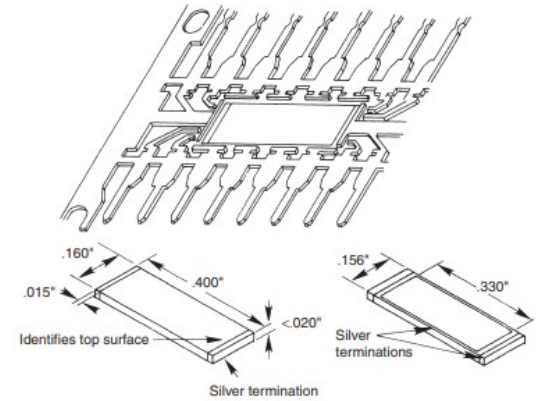
**Figure 5.6** A $V_{cc}$ decoupling capacitor integrated into the IC lead frame. The die can be put on the capacitor, and the whole assembly is ready for conventional bonding/encapsulation (AVX "Bitguard").

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| | |
|---|---|
| Capacitance pin 8 to pin 16 | 0.1 to 0.5 $\mu$fd |
| Normal voltage | –10 V to + 10 V |
| Operating temperature | –55° C to + 125° C |
| Capacitance substrate to pin 16 (optional) | 0.001 to .005 $\mu$fd |
| Typ $\Delta V_{cc}$ for current transient of +20 mA/ns $V_{cc}$ to ground | +0.040 Volts |

**Figure 5.6** A $V_{cc}$ decoupling capacitor integrated into the IC lead frame. The die can be put on the capacitor, and the whole assembly is ready for conventional bonding/encapsulation (AVX "Bitguard").

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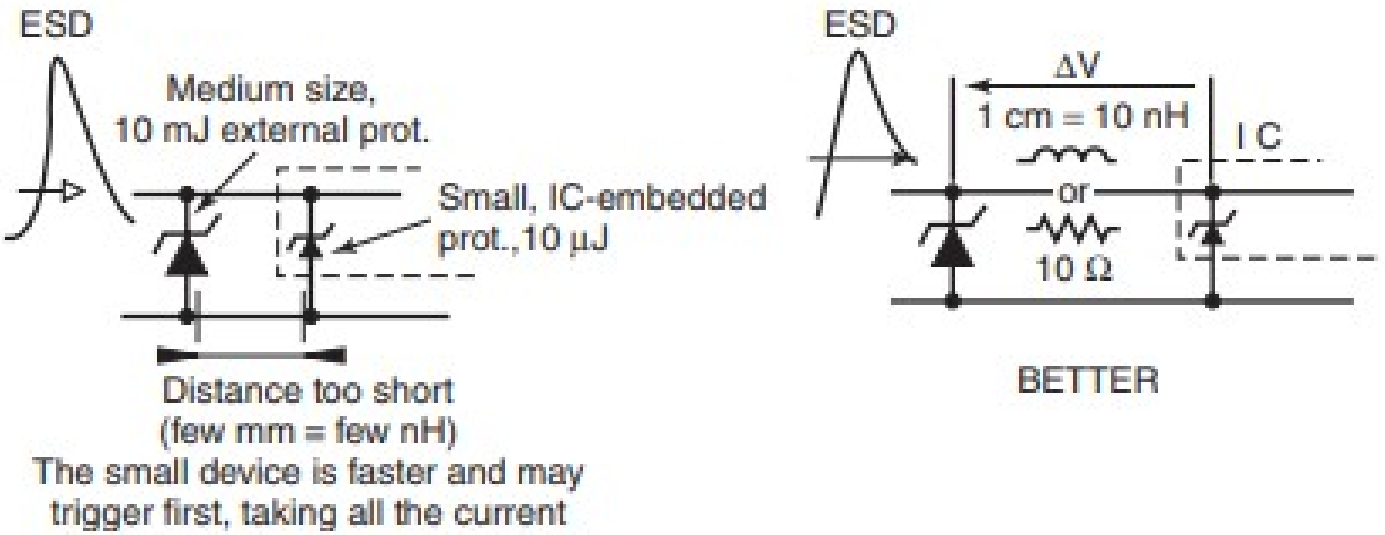



**Figure 5.7** Clamping devices coordination.The voltage drop across a 10-Ω resistance or 10-nH inductance, will raise the voltage seen by the first device (left), forcing its triggering.

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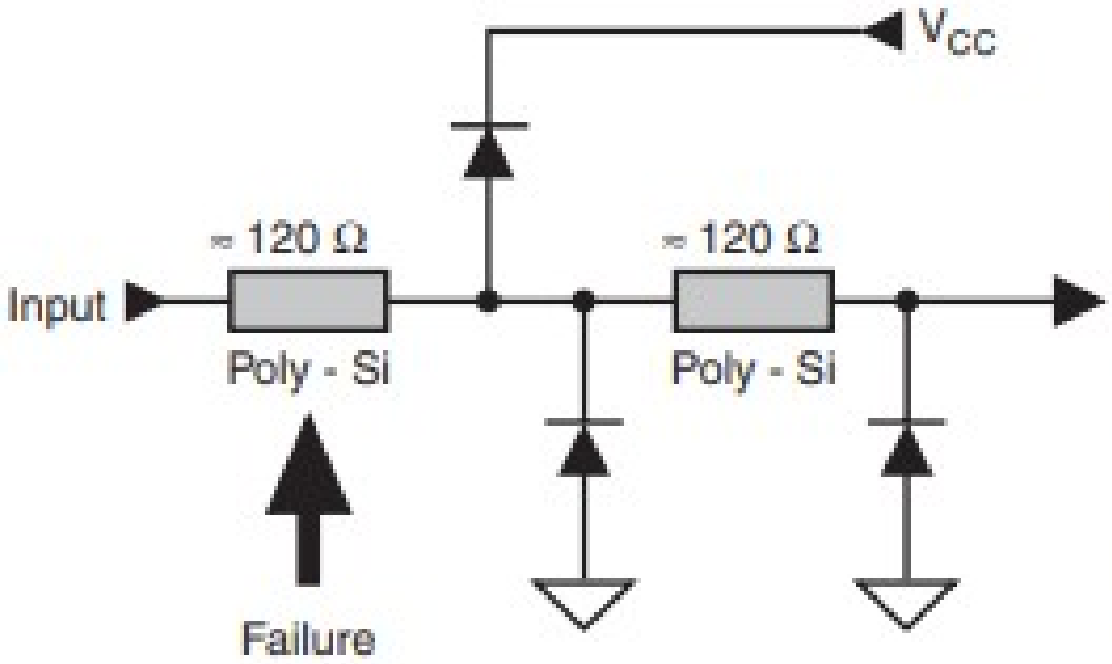



**Figure A.2** Standard HCMOS input protection network. Typical cumulative figures for ESD/HBM failures are: 2050 V (1% defects) to 2700 V (50% defects). More recent processes tend to abandon the poly-Si resistors for the less fragile diffused resistors.

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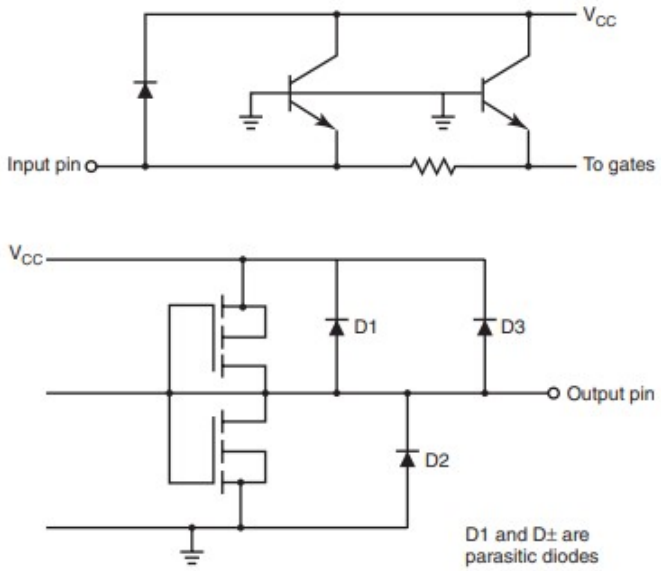



**Figure A.3** High-speed CMOS input and output protection used by Texas Instruments and providing a typical ESD/HBM immunity of 4500 V. The input diode becomes forward-biased for positive pulses exceeding $V_{cc} + 0.5$ V. For negative pulses, the base of the diffused transistor becomes more positive than the input line, turning the transistor ON.

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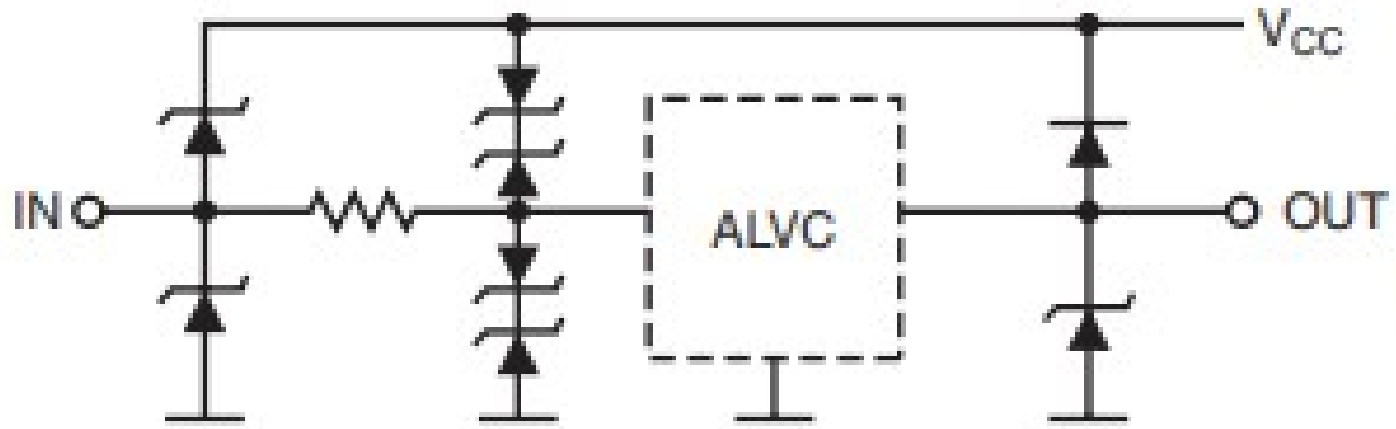



**Figure A.4** ESD protection circuit in advance low-voltage CMOS (ALVC) used by Texas Instruments.

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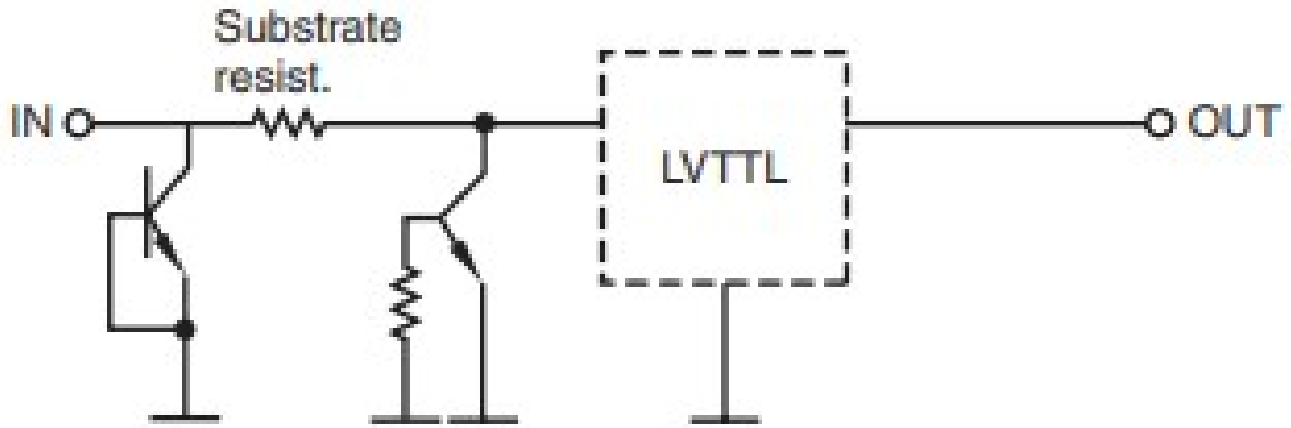


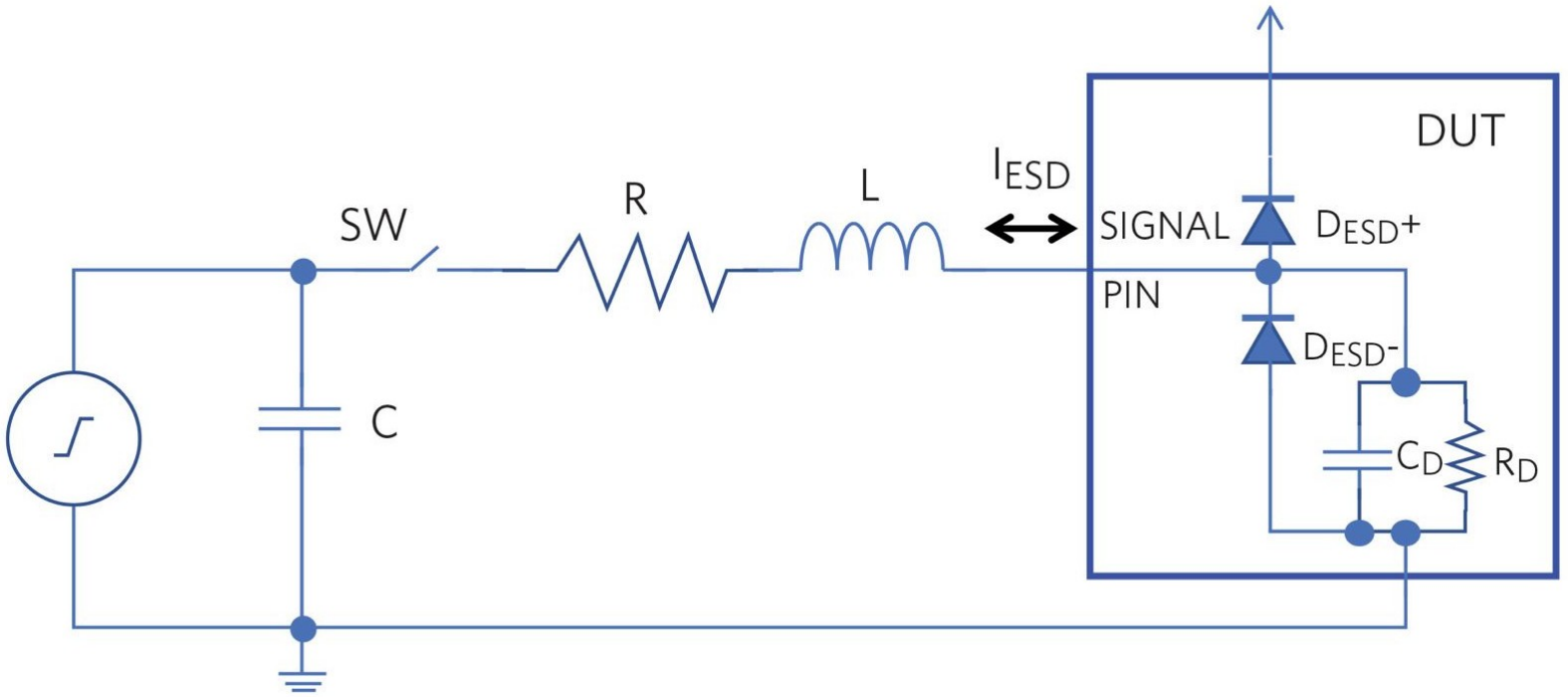

**Figure A.5** Example of cascaded input protection in bipolar low-voltage IC, using a combination of crowbar transistors and a limiting resistor. The output transistors are powerful enough, so no additional protection components are needed.

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**Table 1 – Waveform specification**

| Voltage level (V) | $I_{peak}$ for short, $I_{ps}$ (A) | $I_{peak}$ for 500 Ω $I_{pr}$ (A) | Rise time for short, $t_r$ (ns) | Rise time for 500 Ω $t_{rr}$ (ns) | Decay time for short, $t_d$ (ns) | Maximum ringing current $I_R$ (A) |
|---|---|---|---|---|---|---|
| 125 (optional) | 0,075-0,092 | N/A | 2,0-10 | N/A | 130-170 | 15 % of $I_{ps}$ |
| 250 | 0,15-0,19 | N/A | 2,0-10 | N/A | 130-170 | 15 % of $I_{ps}$ |
| 500 | 0,30-0,37 | N/A | 2,0-10 | N/A | 130-170 | 15 % of $I_{ps}$ |
| 1 000 | 0,60-0,74 | 0,37-0,55 | 2,0-10 | 5,0-25 | 130-170 | 15 % of $I_{ps}$ |
| 2 000 | 1,20-1,48 | N/A | 2,0-10 | N/A | 130-170 | 15 % of $I_{ps}$ |
| 4 000 | 2,40-2,96 | 1,5-2,2 | 2,0-10 | 5,0-25 | 130-170 | 15 % of $I_{ps}$ |
| 8 000 (optional) | 4,80-5,86 | N/A | 2,0-10 | N/A | 130-170 | 15 % of $I_{ps}$ |

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# Lunch Break

**45 minutes**


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ACTIVITY 01 · K1 · A1

# Static Hazard Hunt at NovaSense Incoming Inspection

NovaSense receives MEMS sensor trays beside a label printer, stretch-wrap station and wheeled inspection trolley. Failure returns increased after a layout change, but no one has mapped charge-generation and induction risks.

MISSION

**Identify credible charge sources, exposed ESDS items, isolated conductors and field-coupling paths, then prioritise controls.**

Roles: ESD engineer · incoming-inspection supervisor · quality observer


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ACTIVITY 01

# Scenario questions

## DISCUSS / DECIDE

- Which hazard could damage a device without a visible spark?
- Which metal object could remain charged despite being conductive?
- What evidence would distinguish induction risk from direct contact?

## QUALITY OF REASONING

- Work from the evidence provided.
- State assumptions and missing evidence.
- Use current controlled limits and methods.
- Challenge the first explanation before concluding.


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ACTIVITY 01 · VERIFY

# Evidence and acceptance gate

## EVIDENCE TO SUBMIT

- Annotated hazard map
- Risk-prioritisation worksheet
- Three-control proposal
- Peer-review note

## ACCEPTANCE CRITERIA

✓ At least eight credible hazards are identified

✓ Both direct and field-induced pathways are shown

✓ Top risks have evidence-based ratings

✓ Controls follow the ESD control hierarchy


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ACTIVITY 02 · K1 · A1

# Medical Microchip Failure Triage: ESD, EOS or Process Defect?

A medical-device ASIC passes wafer test but shows intermittent leakage after final assembly. A handler alarm occurred during the lot, yet the power-supply log also shows a transient. Management wants a defensible triage before quarantined product is released.

MISSION

**Build an evidence-based hypothesis that separates HBM/CDM exposure, EOS and non-electrical process causes.**

Roles: reliability engineer · failure-analysis lead · production manager


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ACTIVITY 02

# Scenario questions

## DISCUSS / DECIDE

- Why is a monitor alarm not proof that this device was damaged by ESD?
- Which facts make CDM plausible?
- What would make the latent-failure claim weak?

## QUALITY OF REASONING

- Work from the evidence provided.
- State assumptions and missing evidence.
- Use current controlled limits and methods.
- Challenge the first explanation before concluding.


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ACTIVITY 02 · VERIFY

# Evidence and acceptance gate

## EVIDENCE TO SUBMIT

- Chronology
- Hypothesis matrix
- Containment plan
- Decision-gate briefing

## ACCEPTANCE CRITERIA

✓ ESD, EOS and process alternatives remain explicit

✓ Evidence needed for each hypothesis is stated

✓ Containment protects product and evidence

✓ Conclusions use calibrated language


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RETRIEVAL PRACTICE

# Module 1 recap

1 **Risk**
Charge generation · discharge paths · HBM/MM/CDM · failure modes · business impact

2 **Knowledge**
K1 · A1

3 **Evidence**
Explain the decision and retain traceable records.

4 **Next**
Carry the controls into the following module.


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02

MODULE 2 · K1 · K3 · K4 · A1

# ESD Control Principles, Equipment and Practices

EPA design · grounding and bonding · personnel controls · ionisation · packaging · maintenance


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# The ESD control hierarchy

1. Identify ESDS items and process limits
2. Remove unnecessary charge generators
3. Ground and bond conductors
4. Neutralise essential insulators
5. Shield during storage and transport
6. Verify and improve


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# Electrostatic Protected Area (EPA)

1. **Boundary**
   A defined area with controlled entry and clear visual identification.

2. **Common point**
   A known reference for compatible grounding and bonding.

3. **Approved items**
   Only qualified materials, tools and equipment inside.

4. **Behaviour**
   Handling rules, training and compliance verification are enforced.


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# Grounding versus bonding

## PRINCIPLE / USE

- Grounding connects a conductor to a verified reference.
- Personnel and equipment paths must be compatible with safety requirements.

## RISK / DECISION

- Bonding connects conductors so they remain at substantially the same potential.
- Bond isolated metal objects even when they are not intentionally energised.


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# Common point ground architecture

1. Facilities/equipment ground verified
2. EPA common point established
3. Mat and fixtures bonded
4. Personnel grounding connected
5. Continuity monitored and recorded


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# Personnel grounding choices

1

## Seated work

Wrist strap with tested cord and verified connection.

2

## Standing/mobile

Foot grounders or qualified footwear/flooring system.

3

## Continuous monitoring

Real-time alarm when the selected path leaves limits.


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# Wrist-strap control loop

1. Inspect band and cord
2. Test before use
3. Connect to approved point
4. Maintain skin contact
5. Respond immediately to alarm/failure


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# Footwear and flooring as a system

## PRINCIPLE / USE

- Foot grounders/ESD footwear provide the personnel interface.
- Correct wear and both-foot coverage matter.

## RISK / DECISION

- Qualified flooring completes the path.
- Contamination, wax, humidity and wear can change performance.


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# Worksurface and tool control

1. **Dissipative mat**
   Provides controlled charge decay and a defined work zone.

2. **Grounded fixtures**
   Prevents isolated conductive metal.

3. **ESD-safe tools**
   Tweezers, brushes and dispensers selected for the process.

4. **Housekeeping**
   Cleaners and residues must not defeat electrical properties.


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# Essential insulators

1. Identify the process need
2. Measure field at the actual use distance
3. Increase separation where possible
4. Apply ionisation or shielding
5. Verify after changes


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# Ionisation

## PRINCIPLE / USE

- Balanced positive and negative ions neutralise charges on insulators and isolated conductors.
- Use where grounding is ineffective or impossible.

## RISK / DECISION

- Verify offset/balance and decay performance at the work location.
- Airflow, distance, contamination and emitter maintenance affect results.


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# Electrostatic field shielding

1

## Conductive enclosure

Redistributes external charge and protects contents when closed.

2

## Shielding bag

Provides a protective envelope for transport outside the EPA.

3

## Continuity

Seams, closures and openings determine actual performance.


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# Packaging functions are different

1 **Low charging**
Limits triboelectric generation during contact/separation.

2 **Dissipative**
Moves charge in a controlled way.

3 **Conductive**
Rapidly redistributes charge; may need grounding.

4 **Shielding**
Attenuates discharge current and electrostatic fields around contents.


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# Handling ESDS items

1. Confirm EPA readiness
2. Ground the handler
3. Open shielding package inside EPA
4. Handle by body/edges
5. Return to protective packaging before leaving


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# Garments, gloves and finger cots

## PRINCIPLE / USE

- Garments control charge from clothing and help contain personal items.
- Select design and grounding method for the process.

## RISK / DECISION

- Gloves/finger cots may support contamination control but are not automatically ESD-safe.
- Qualification and correct use are required.


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# Carts, racks and material movement

1

## Conductive wheels

Maintain the intended path only when floor contact is suitable.

2

## No loose insulators

Remove ordinary plastic wrap and foam near exposed ESDS items.

3

## Covered transport

Use closed qualified containers outside EPA boundaries.

4

## Route control

Avoid high-charging areas and unmanaged handoffs.


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# Maintenance is part of control

1. Define owner and interval
2. Inspect and clean
3. Test with calibrated equipment
4. Record result and disposition
5. Trend failures and revise interval


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# Common failure patterns

1. **Disconnected strap**
   Cord attached to an unverified point or worn over clothing.

2. **Open bag**
   Shielding package left open or folded incorrectly.

3. **Insulator creep**
   Tape, foam, trays or paperwork introduced without review.

4. **Bad response**
   Alarm bypassed, failed tester ignored or record completed retrospectively.


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# Supervisor’s EPA readiness review

1

## People

Training current; grounding tested; behaviour observed.

2

## Place

Boundary, common point, floor, mat and ioniser verified.

3

## Product

Sensitivity known; packaging and process controls matched.


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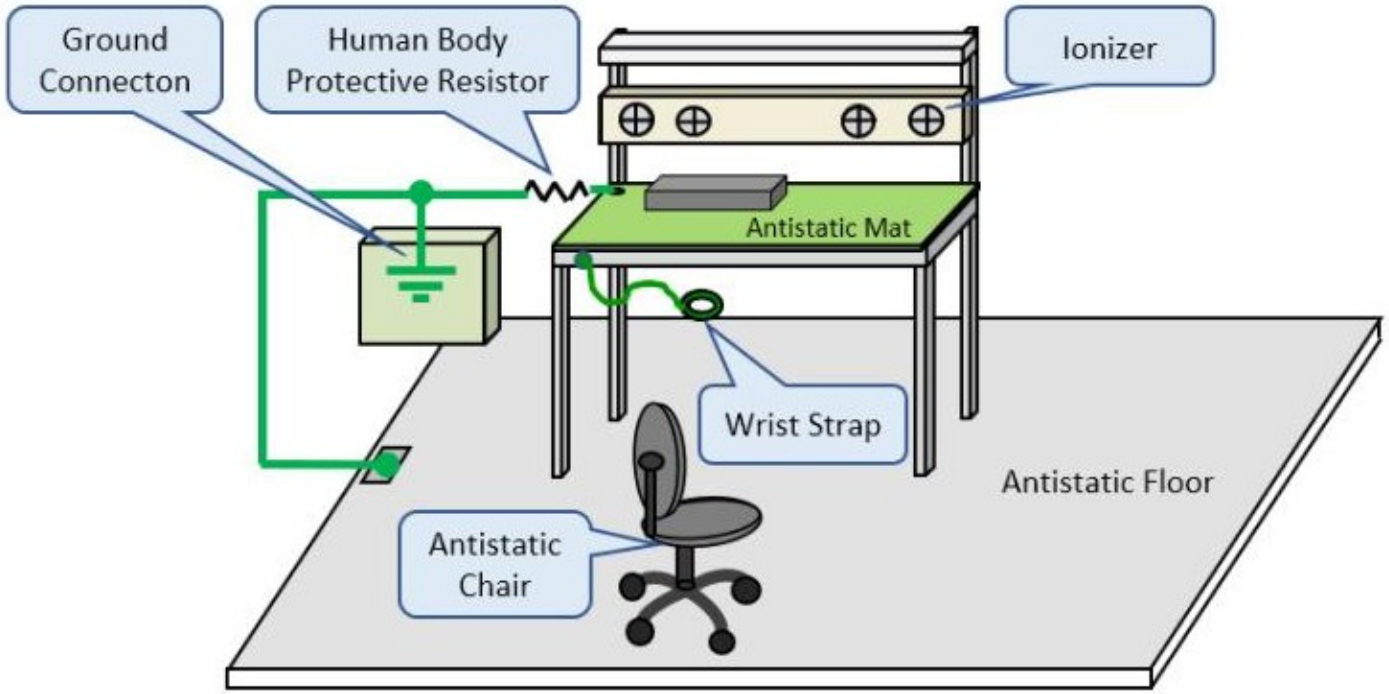



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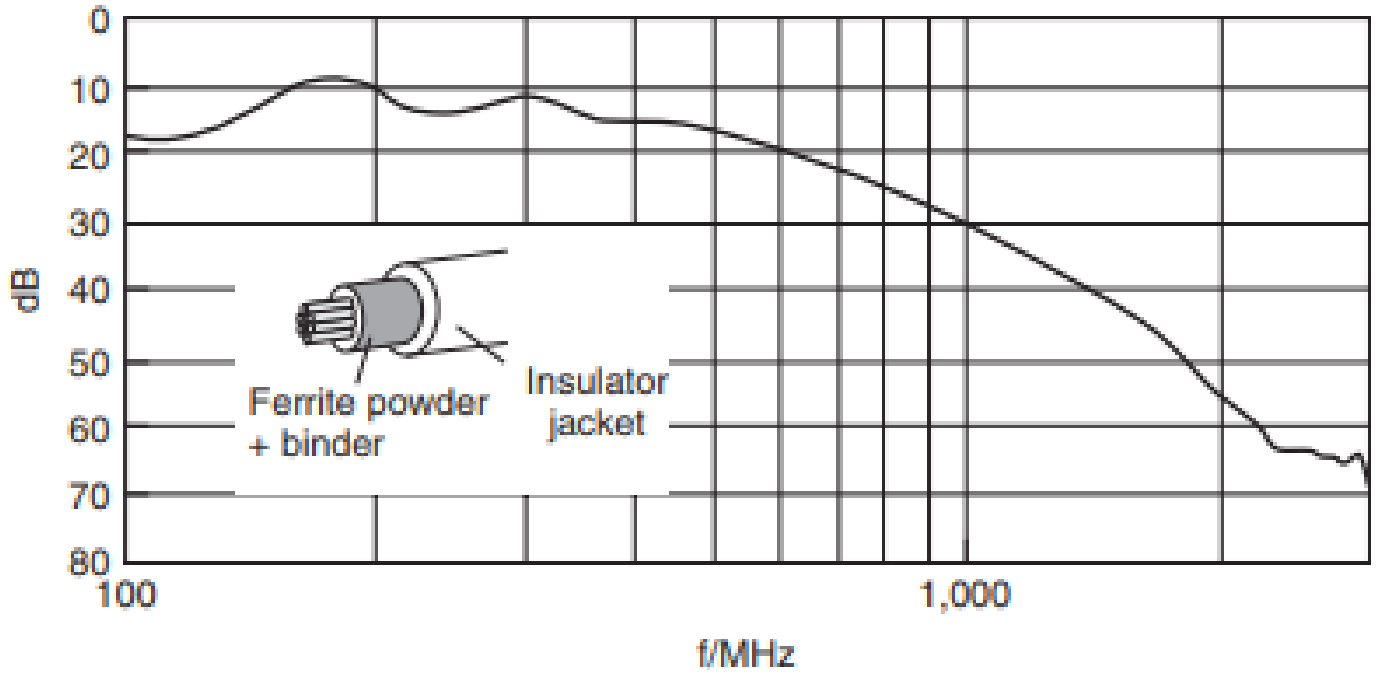



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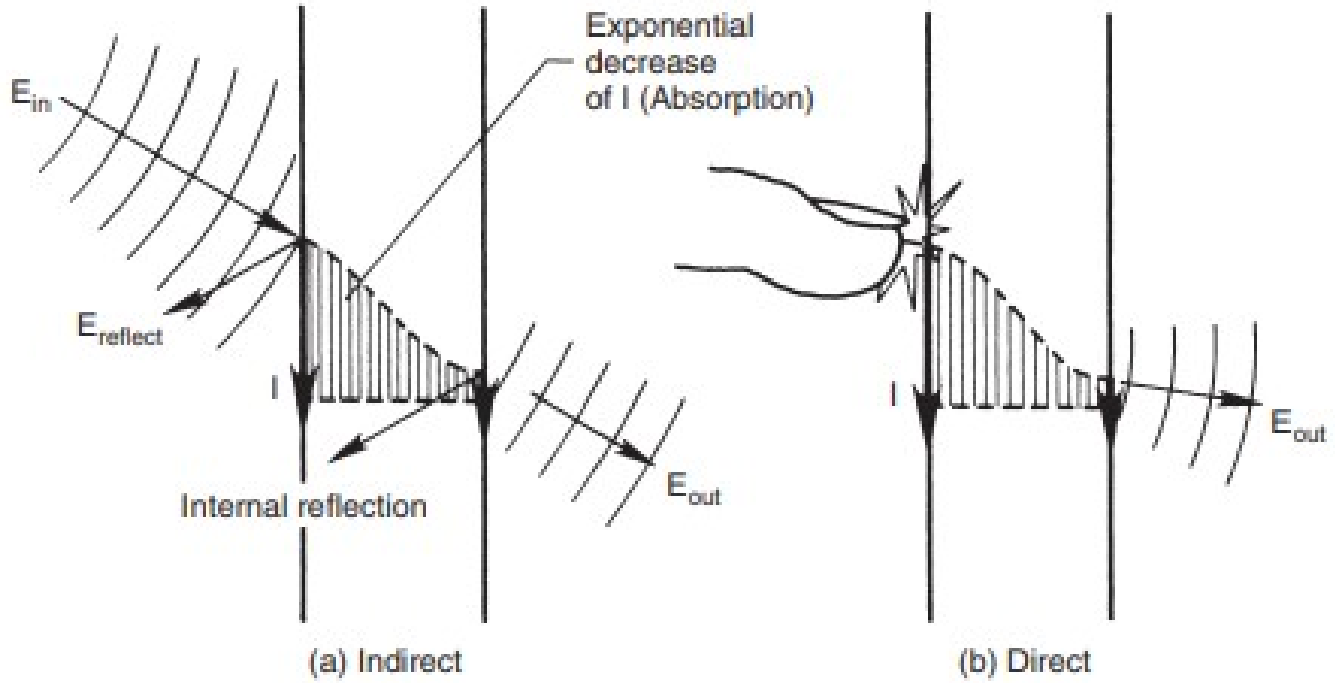



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**Table 5.2** Absorption Loss in dB of Several Metals[a]

| Thickness | Copper | | | Aluminum | | | Zinc | | | Steel* | | | Nickel* $\sigma_r = 0.2$ | | Copper Paint (nonhomogeneous metal) $\sigma_r = 0.04$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| (mm) | 0.01 | 0.1 | 1 | 0.01 | 0.1 | 1 | 0.01 | 0.1 | 1 | 0.01 | 0.1 | 1 | .01 | 0.1 | 0.05 mm (2 mil) |
| 30 MHz | 7 | 70 | 700 | 5.2 | 52 | 520 | 4 | 40 | 400 | 3 | 28 | 280 | 3 | 31 | 7 |
| 100 MHz | 13 | 130 | >1000 | 9.5 | 95 | 950 | 7 | 72 | 720 | 5 | 50 | 500 | 6 | 58 | 13 |
| 300 MHz | 22 | 220 | >1000 | 17 | 170 | >1000 | 12 | 125 | >1000 | 9 | 88 | 880 | 10 | 98 | 22 |

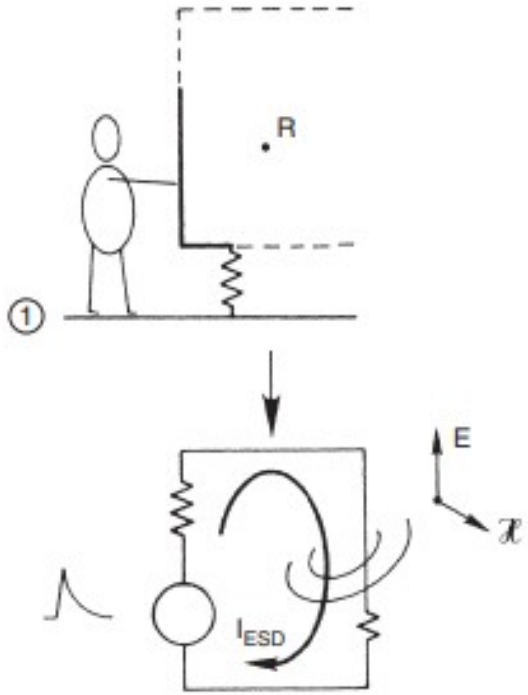
[a]The shielding effectiveness against ESD will be *at least* equal to this plus some reflection. For steel and nickel, the values shown take into account the fall-off of their magnetic permeability above ≈ few MHz.

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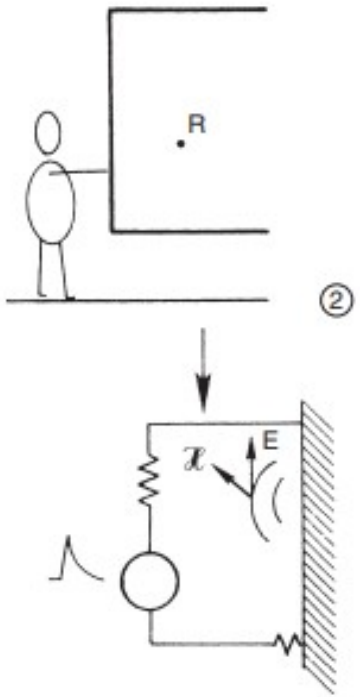



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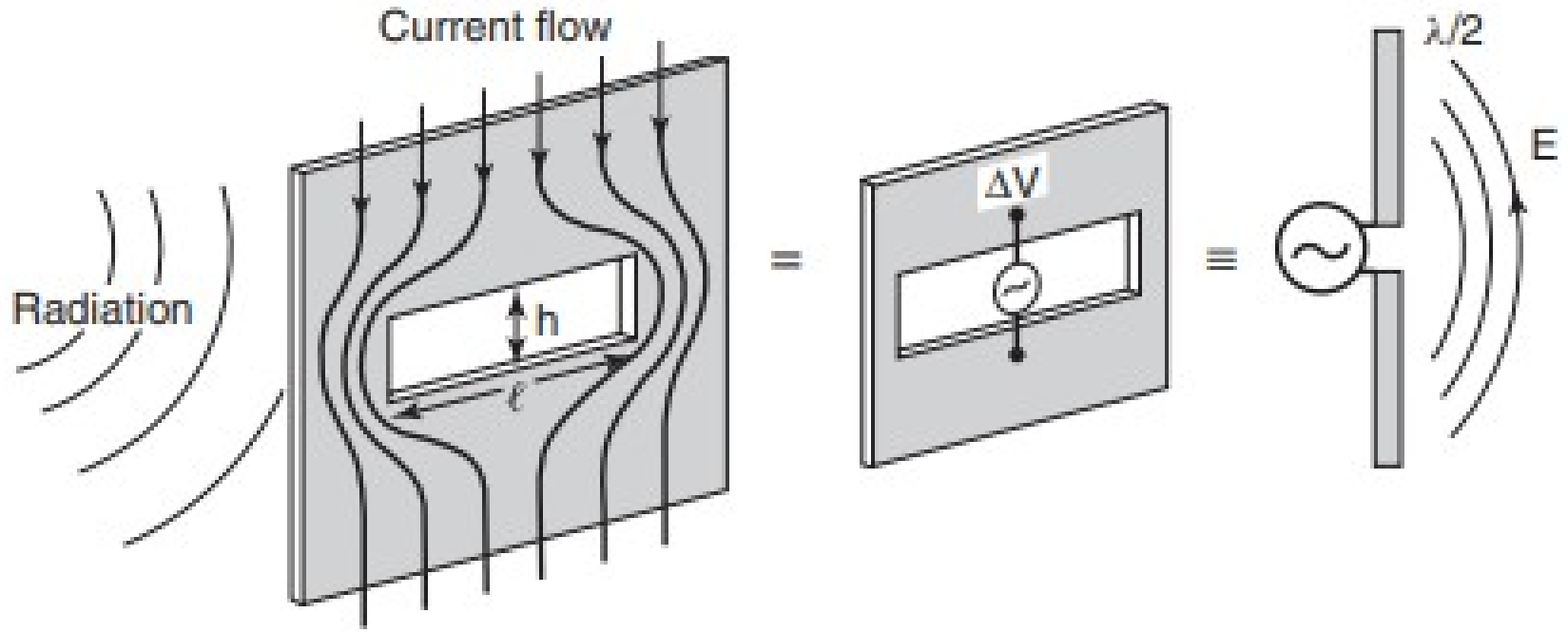



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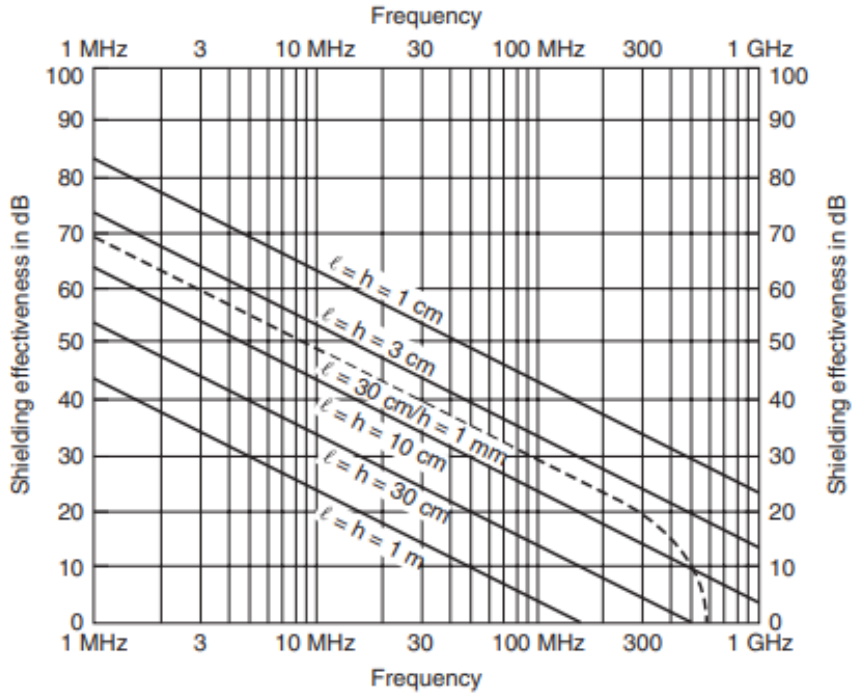



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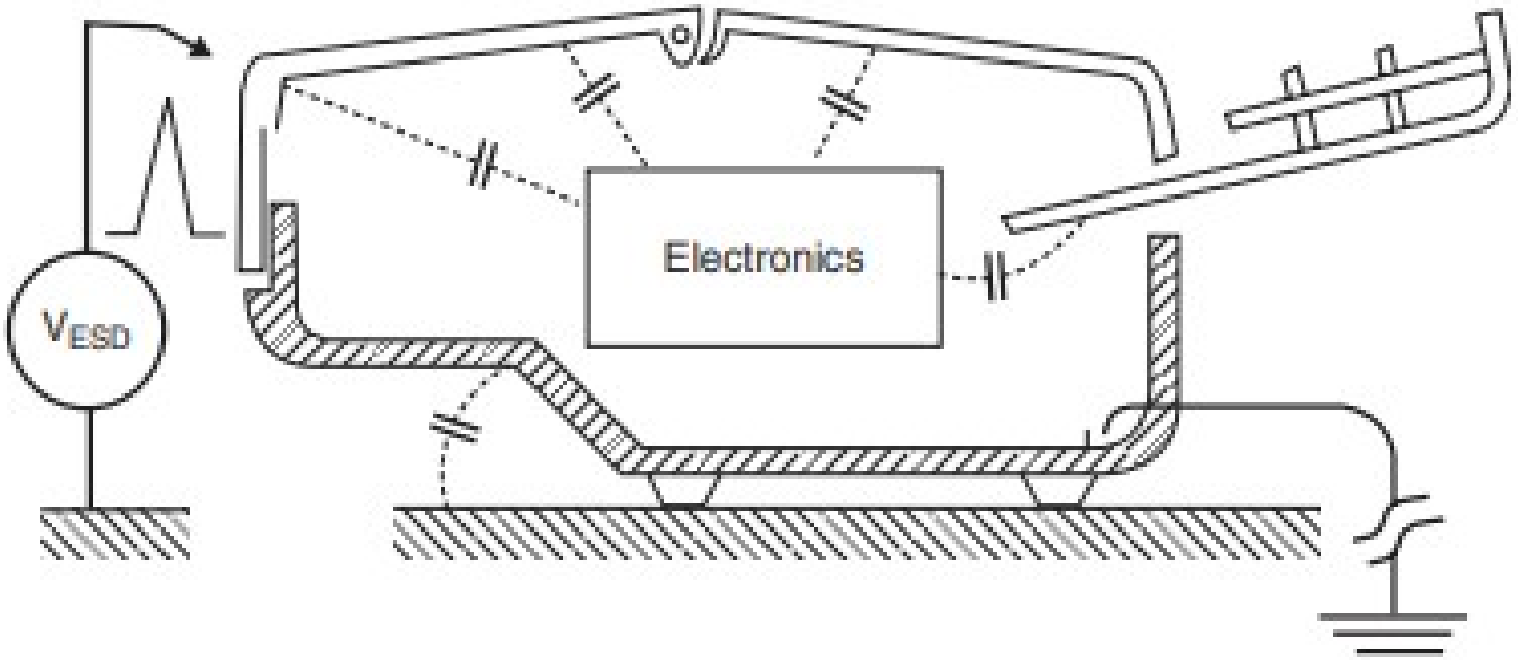



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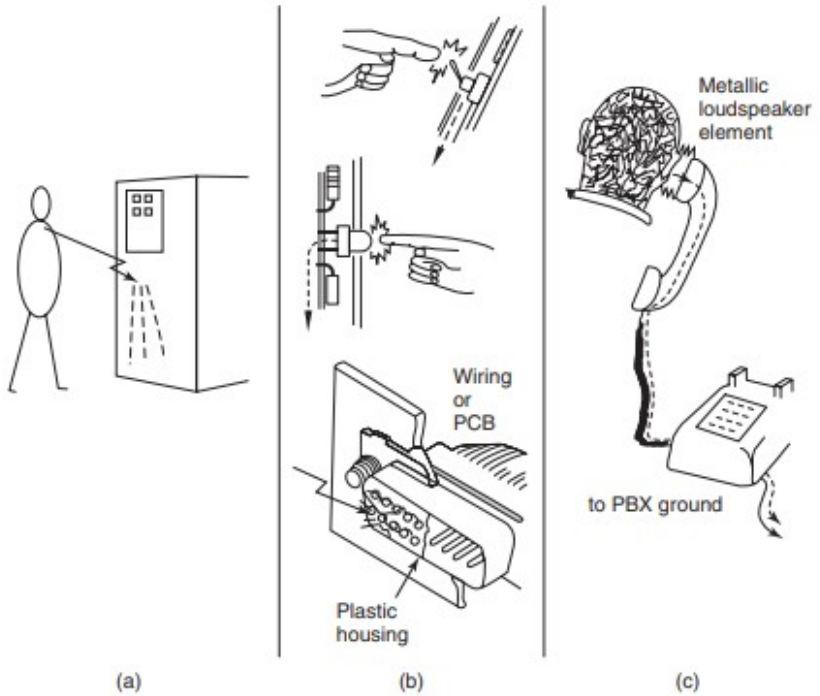


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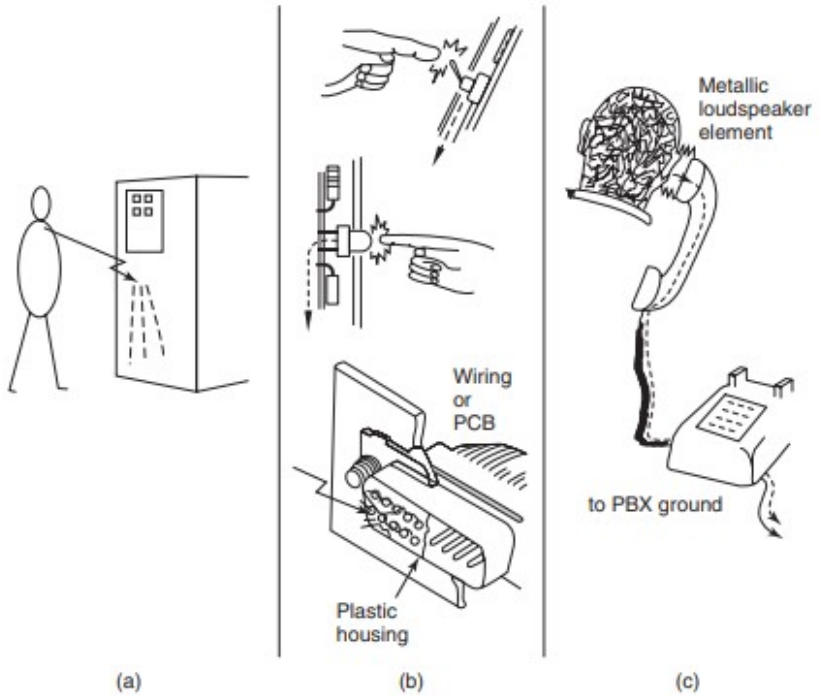



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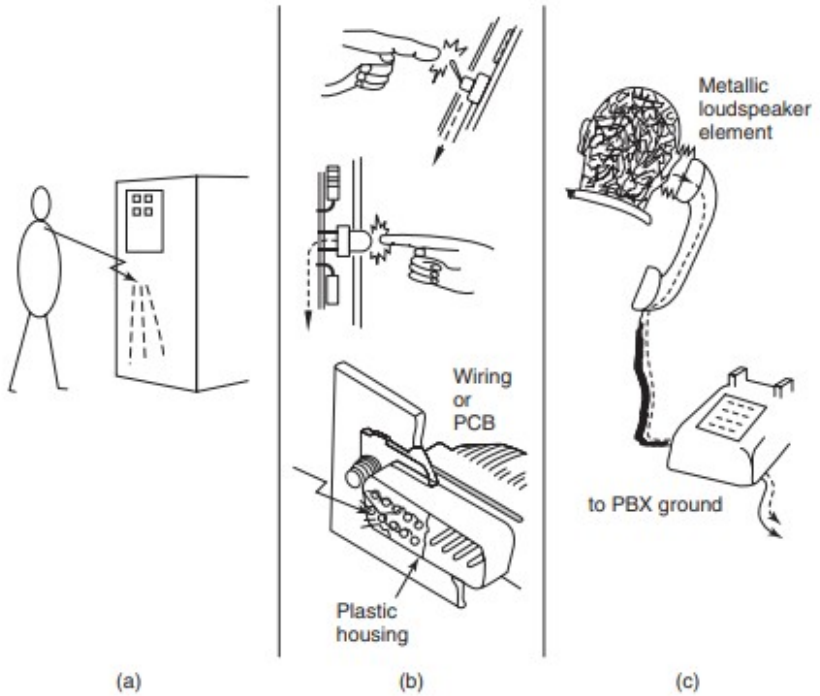



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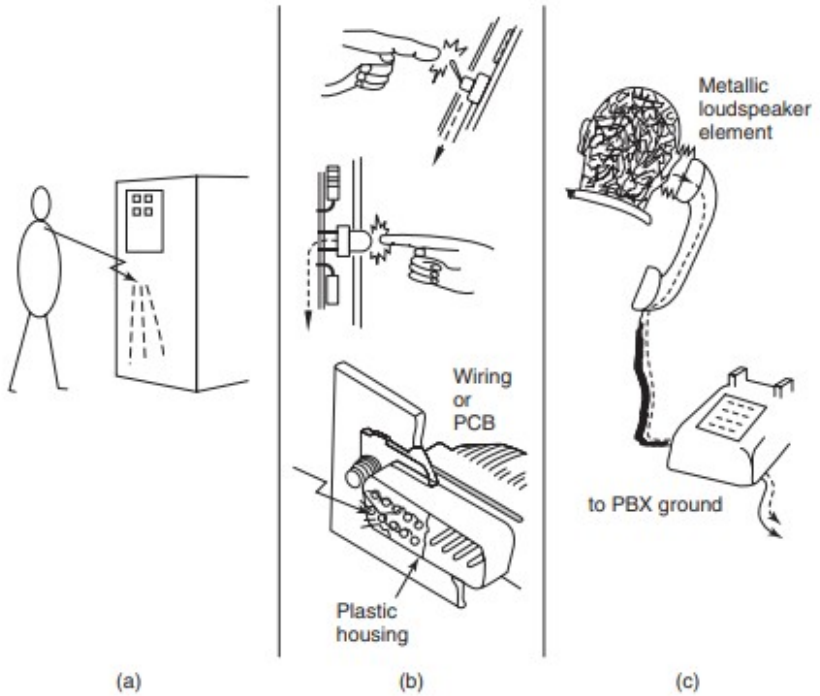



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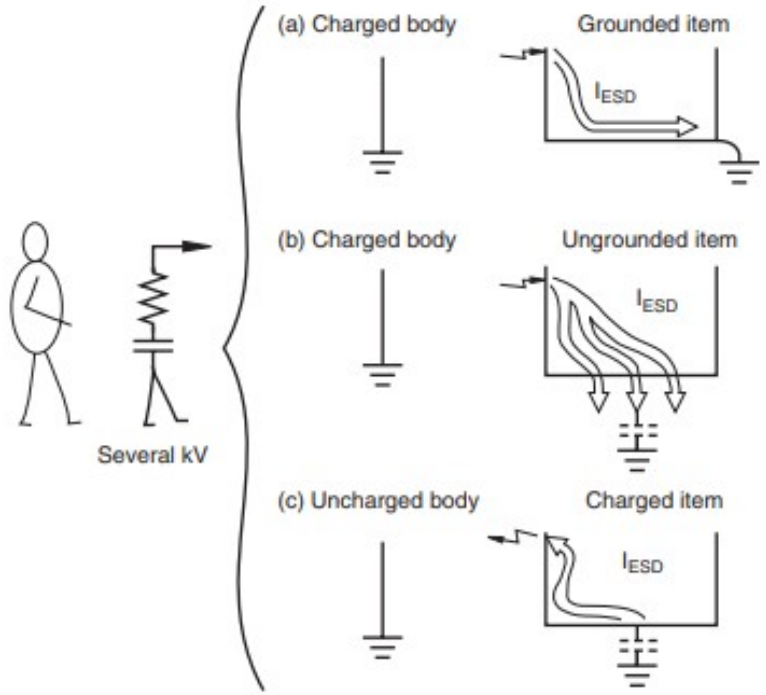



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ACTIVITY 03 · K3 · K4 · A1

# Design and Commission an EPA for Orion Optics

Orion Optics is moving laser-diode assembly into a small shared lab. The proposed bench has a mat and wrist strap but no documented boundary, common point, insulator policy or readiness test.

MISSION

**Design an EPA that controls people, place, product and process, then define commissioning evidence.**

Roles: ESD engineer · facilities engineer · lab manager


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ACTIVITY 03

# Scenario questions

## DISCUSS / DECIDE

- What makes the area an EPA rather than simply a bench with a mat?
- Which item cannot be made safe by grounding alone?
- What evidence authorises production start?

## QUALITY OF REASONING

- Work from the evidence provided.
- State assumptions and missing evidence.
- Use current controlled limits and methods.
- Challenge the first explanation before concluding.


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ACTIVITY 03 · VERIFY

# Evidence and acceptance gate

## EVIDENCE TO SUBMIT

- EPA layout
- Grounding/bonding diagram
- Insulator-control register
- Commissioning plan and readiness certificate

## ACCEPTANCE CRITERIA

✓ Boundary and product flow are unambiguous

✓ All conductors have a resolved status

✓ Essential insulators have controls

✓ Commissioning includes method, limit, owner and response


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ACTIVITY 04 · K3 · K4 · A1

# Select Controls for the Aster Robotics Handling Cell

A robot transfers packaged ICs from plastic tubes to a grounded nest. Operators also perform manual rework. The team proposes more wrist straps, but field readings near the tube exit remain high.

MISSION

**Match grounding, ionisation, shielding, handling and monitoring controls to both human and automated exposure pathways.**

Roles: automation engineer · ESD specialist · maintenance supervisor


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ACTIVITY 04

# Scenario questions

## DISCUSS / DECIDE

- Why will adding wrist straps not solve the tube-exit field?
- Which controls address CDM risk?
- What should happen when an ioniser check fails?

## QUALITY OF REASONING

- Work from the evidence provided.
- State assumptions and missing evidence.
- Use current controlled limits and methods.
- Challenge the first explanation before concluding.


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ACTIVITY 04 · VERIFY

# Evidence and acceptance gate

## EVIDENCE TO SUBMIT

- Exposure-path map
- Control-selection matrix
- Monitoring/stop-condition plan
- Maintenance plan

## ACCEPTANCE CRITERIA

✓ Controls cover human and automated pathways

✓ Insulators are addressed by an appropriate method

✓ Monitoring has explicit response actions

✓ Residual risk is stated


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RETRIEVAL PRACTICE

# Module 2 recap

1

## Risk

EPA design · grounding and bonding · personnel controls · ionisation · packaging · maintenance

2

## Knowledge

K1 · K3 · K4 · A1

3

## Evidence

Explain the decision and retain traceable records.

4

## Next

Carry the controls into the following module.


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# Day 1 Complete

**Return on time for Day 2 digital attendance**


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03

MODULE 3 · K2 · K3 · A2 · A3

# ESD Standards, Testing and Auditing

Control programme structure · measurements · compliance verification · evidence · performance metrics

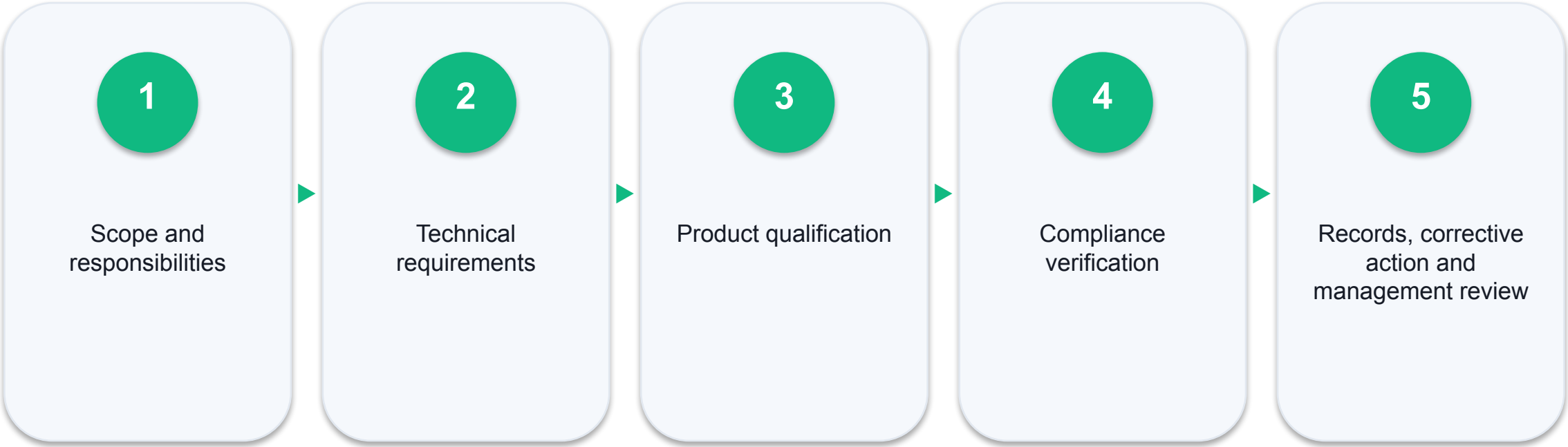

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# ANSI/ESD S20.20 and IEC 61340-5-1 structure organisational controls for protecting ESDS items.

Use the current controlled copy, customer requirements and product sensitivity data for exact limits and methods.


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# Control programme architecture





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# Product qualification versus compliance verification

## PRINCIPLE / USE

- Qualification demonstrates that an ESD control item/type is capable of meeting specified requirements.
- Evidence may include supplier data or internal testing.

## RISK / DECISION

- Compliance verification checks that installed controls remain effective in routine use.
- It uses defined methods, frequencies, limits and response actions.


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# Instrument-to-risk map

1

## Resistance meter

Worksurfaces, floors, garments, packaging and ground paths.

2

## Personnel tester

Wrist straps, footwear and person/footwear/floor systems.

3

## Field meter

Electrostatic field and charged-object screening.

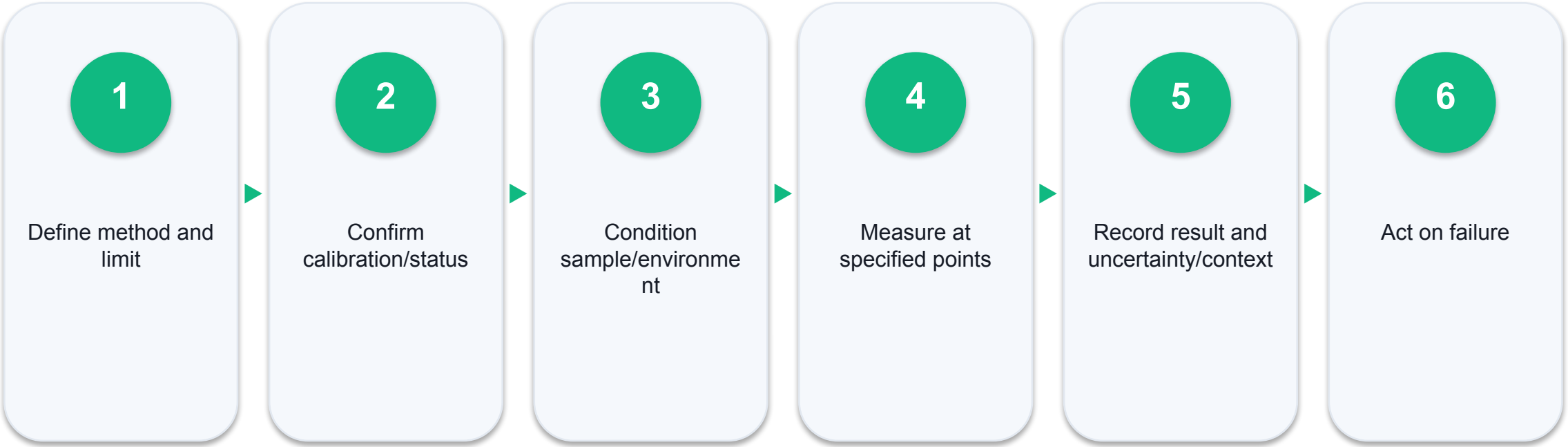
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## Charge plate monitor

Ioniser decay and offset/balance performance.


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# Measurement discipline





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# Wrist strap and footwear checks

## PRINCIPLE / USE

- Daily/user checks screen the assembled personal grounding path before work.
- Failure requires immediate stop, retest and disposition.

## RISK / DECISION

- Periodic technical checks verify components, testers and flooring systems.
- Trend repeated failures by user, device, location and shift.


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# Surface and point-to-ground resistance

1

## Point-to-point

Characterises the path across a surface/material.

2

## Point-to-ground

Characterises the installed path to the designated ground.

3

## Context

Electrodes, voltage, dwell, humidity and cleanliness influence the reading.


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# Ioniser performance checks

1. Inspect placement and airflow
2. Warm up per procedure
3. Measure positive/negative decay
4. Measure offset/balance
5. Clean or adjust
6. Repeat and record


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# Audit evidence pyramid

1 **Observe**
Actual behaviour at the workstation.

2 **Measure**
Current test results from calibrated instruments.

3 **Interview**
Personnel can explain risk and response.

4 **Records**
Training, qualification, maintenance, failures and CAPA are traceable.


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# Risk-based audit sampling

## PRINCIPLE / USE

- Increase attention where products are most sensitive, exposure is high or process changes are recent.
- Sample across shifts, people and workstations.

## RISK / DECISION

- Do not let a checklist replace observation and measurement.
- A single passing sample does not prove the entire system.


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# Nonconformity grading

1

## Critical

Immediate product/safety exposure; contain and escalate.

2

## Major

Systemic or repeated breakdown; formal CAPA and verification.

3

## Minor/observation

Local weakness or improvement opportunity; track proportionately.


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# Corrective-action loop





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# Useful ESD performance metrics

1

## Leading

Test completion, first-pass pass rate, overdue calibration, training currency.

2

## Process

EPA audit score, insulator findings, alarm response time.

3

## Lagging

ESD-attributed escapes, scrap, rework and field returns.

4

## Learning

Repeat causes, CAPA recurrence and time-to-effective closure.


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# Management review questions

1

## Effective?

Are controls reducing exposure and failures?

2

## Adequate?

Do changes in products/processes require new controls?

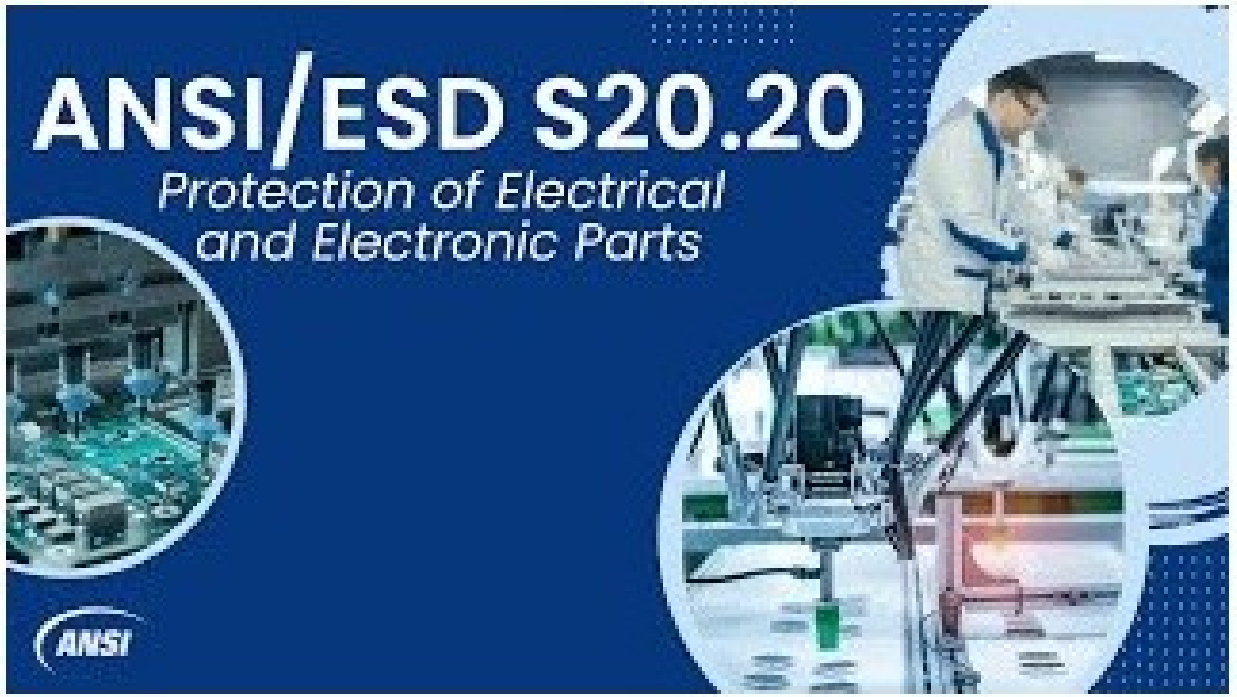
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## Sustained?

Are verification, training and CAPA completed on time?


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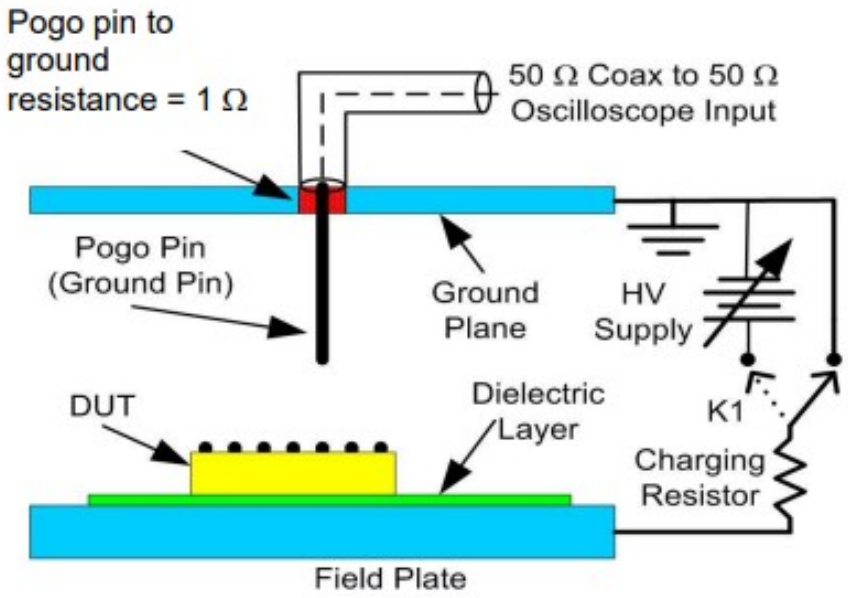



*Figure 1: Simplified CDM Tester Hardware Schematic*

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**Table 3. CDM ESDS Device Classification Levels**

| Classification Level (see Note 1) | Classification Test Condition (in Volts) (See Note 2) |
|---|---|
| C0a | < 125 |
| C0b | 125 to < 250 |
| C1 | 250 to < 500 |
| C2a | 500 to < 750 |
| C2b | 750 to < 1000 |
| C3 | ≥ 1000 (see Note 3) |

NOTE 1: Use the "C" prefix to indicate a CDM classification level.

NOTE 2: The Classification Test Condition is not equivalent to the actual set voltage of the tester. Please see Section 6.5.1 and Annex H for further details.

NOTE 3: For Test Conditions above 1,000 volts, depending on geometry of the device package, corona effects may limit the actual pre-discharge voltage and discharge current.

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**Table 4 – Specification for CDM Tester Verification Modules (Metal Discs)**

| Disk | Small | Large |
|---|---|---|
| Diameter mm (inches) | 8.89 ± 0.127 (0.350 ± 0.005) | 25.4 ± 0.127 (1.000 ± 0.005) |
| Thickness mm (inches) | 1.27 ± 0.05 (0.05 ± 0.002) | 1.27 ± 0.05 (0.050 ± 0.002) |
| Surface flatness variation mm (inches) | ≤ 0.127 (0.005) | ≤ 0.127 (0.005) |
| Capacitance at 1 MHz (pF) | 6.8 ± 5% | 55 ± 5% |

Old PPT slide 210 · picture 1.

Imported in full for legacy traceability. Current controlled standards and procedures govern operational decisions.


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ACTIVITY 05 · K2 · K3 · A2

# Execute a Compliance Verification Round

Meridian Electronics has four EPAs on two shifts. Recent records show perfect results but several tester failures were repaired without documented product disposition. You must execute a verification round using a simulated data pack.

MISSION

**Plan, execute and record traceable checks for personnel grounding, resistance, electrostatic field and ioniser performance.**

Roles: ESD auditor · area owner · instrument custodian


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ACTIVITY 05

# Scenario questions

## DISCUSS / DECIDE

- Why is a retest pass insufficient without investigating the first failure?
- Which contextual data makes a resistance reading defensible?
- Where should ioniser performance be checked?

## QUALITY OF REASONING

- Work from the evidence provided.
- State assumptions and missing evidence.
- Use current controlled limits and methods.
- Challenge the first explanation before concluding.


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ACTIVITY 05 · VERIFY

# Evidence and acceptance gate

## EVIDENCE TO SUBMIT

- Completed measurement log
- Failure/disposition records
- Metric calculation
- Signed evidence package

## ACCEPTANCE CRITERIA

✓ Every result includes method, units and conditions

✓ Instrument status is verified

✓ First-pass failures remain visible

✓ Product impact and response are documented


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ACTIVITY 06 · K2 · A2 · A3

# Standards-Based EPA Audit at Helix Assembly

Helix Assembly claims alignment to ANSI/ESD S20.20 and IEC 61340-5-1. A new customer requests evidence that controls are implemented, verified and improved—not merely purchased.

MISSION

**Conduct a risk-based audit using observation, measurement, interview and records, then issue graded findings and corrective-action requests.**

Roles: lead auditor · ESD programme coordinator · customer observer


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ACTIVITY 06

# Scenario questions

## DISCUSS / DECIDE

- What evidence shows a control is sustained rather than merely installed?
- Why must the exact controlled standard govern numerical limits?
- What makes a finding actionable?

## QUALITY OF REASONING

- Work from the evidence provided.
- State assumptions and missing evidence.
- Use current controlled limits and methods.
- Challenge the first explanation before concluding.


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ACTIVITY 06 · VERIFY

# Evidence and acceptance gate

## EVIDENCE TO SUBMIT

- Audit plan
- Completed checklist
- Evidence register
- Graded findings and follow-up plan

## ACCEPTANCE CRITERIA

✓ Sampling is risk-based

✓ Findings cite objective evidence

✓ Critical exposure triggers containment

✓ Follow-up tests effectiveness, not only completion


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RETRIEVAL PRACTICE

# Module 3 recap

1 **Risk**
Control programme structure · measurements · compliance verification · evidence · performance metrics

2 **Knowledge**
K2 · K3 · A2 · A3

3 **Evidence**
Explain the decision and retain traceable records.

4 **Next**
Carry the controls into the following module.


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# Afternoon Break

**10 minutes**


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04

MODULE 4 · K4 · A2 · A3

# Training, Awareness and Incident Improvement

Competence · behaviour · reporting · root-cause analysis · CAPA · management ownership


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# Competence is role-specific

1

## Operators

Recognise ESDS items, enter EPA correctly and handle/pack safely.

2

## Technicians

Set up, test, troubleshoot and maintain controls.

3

## Engineers

Define limits, qualify controls and investigate events.

4

## Managers

Provide resources, review metrics and remove systemic barriers.


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# From awareness to reliable behaviour

1. Explain the invisible risk
2. Demonstrate the correct method
3. Practise with feedback
4. Observe on the job
5. Reinforce and requalify


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# ESD champions and supervision

1

## Coach

Give immediate, specific feedback at the workstation.

2

## Escalate

Stop unsafe handling and trigger containment when required.

3

## Learn

Share incident lessons without normalising blame.


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# Recognising a possible ESD incident

1

## Product symptom

Sudden failure, parametric shift, intermittent behaviour or unexplained fallout.

2

## Control signal

Monitor alarm, failed tester, ioniser drift or broken ground path.

3

## Exposure clue

Unprotected handling, ordinary plastic, low humidity or process change.

4

## Caution

Symptoms are not proof; preserve evidence for diagnosis.


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# Immediate response

1. Stop and protect the process
2. Segregate potentially affected product
3. Preserve settings, logs and samples
4. Notify defined roles
5. Begin evidence-based triage


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# Incident evidence set

1. **What/where/when**
   Product, lot, station, timestamp and sequence.

2. **Environment**
   Humidity, temperature, nearby motion and insulators.

3. **Controls**
   Tester logs, monitor alarms, grounding and ioniser status.

4. **People/process**
   Who handled it, procedure used and recent changes.


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# Root-cause analysis without blame

## PRINCIPLE / USE

- Ask why the system allowed exposure: design, maintenance, method, material, measurement or management.
- Distinguish root, contributing and detection causes.

## RISK / DECISION

- Avoid stopping at 'operator error'.
- A repeated behaviour usually reflects unclear standards, weak design, workload or ineffective supervision.


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# Selecting corrective actions

1. Remove generator/exposure
2. Engineer a reliable ground or enclosure
3. Add detection/monitoring
4. Standardise work and response
5. Train and verify competence
6. Measure sustained effectiveness


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# Communicating ESD risk

1

## Frontline

Show the exact handling risk and the stop-work response.

2

## Engineering

Use measurements, failure evidence and process capability.

3

## Management

Translate risk into yield, customer, compliance and continuity impact.


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# A sustainable ESD culture

1

## Visible ownership

Named programme coordinator and local owners.

2

## Just reporting

Near misses and failures can be reported without concealment.

3

## Closed loop

Findings become actions, verified changes and updated standards.

4

## Learning cadence

Metrics, incidents and changes are reviewed regularly.


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# Competency certification

## PRINCIPLE / USE

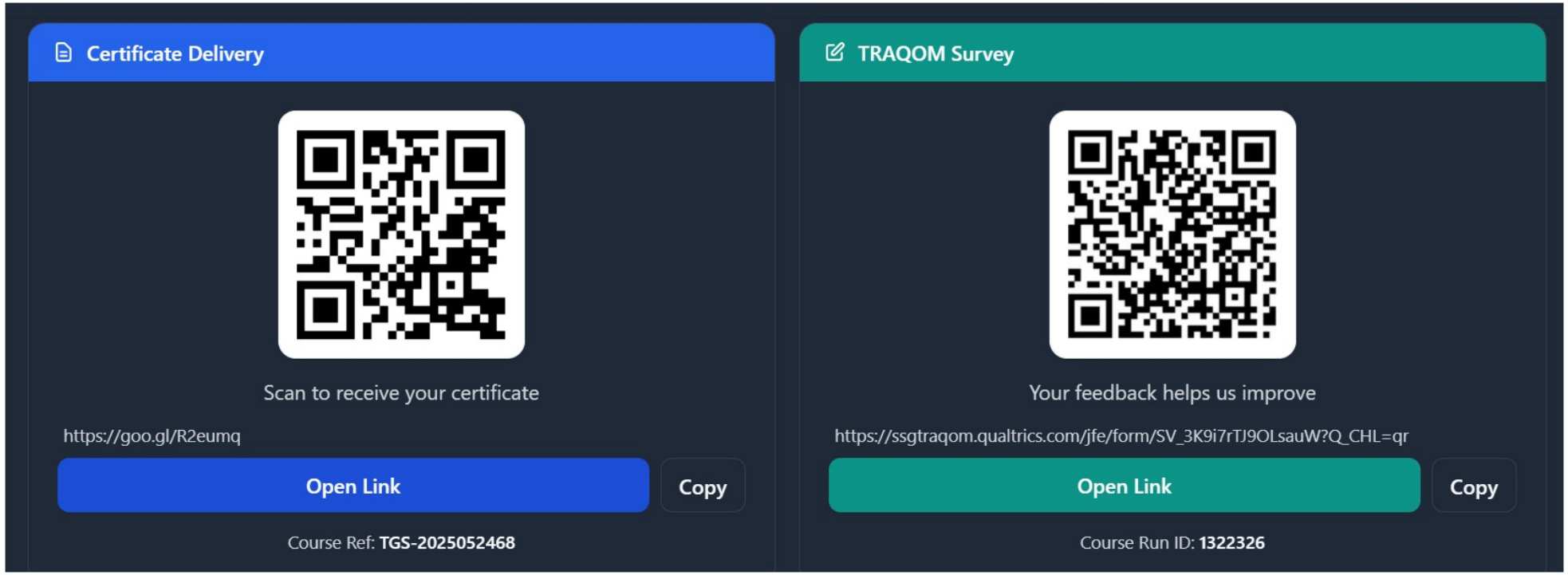
- Internal authorisation should match the tasks a person performs.
- Use knowledge, demonstration and observed-work evidence.

## RISK / DECISION

- External credentials may support development but do not replace site-specific qualification.
- Reassess after major changes, lapse or repeated nonconformance.


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# Imported legacy visual reference




Old PPT slide 267 · picture 1. Historical QR/screen capture only—use the live LMS.

**Imported in full for legacy traceability. Current controlled standards and procedures govern operational decisions.**


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ACTIVITY 07 · K4 · A2 · A3

# Investigate the PulseMed ESD Incident

PulseMed sees intermittent failures in a safety-relevant control board after a temporary operator used ordinary bubble wrap near final test. A monitor alarm was acknowledged but production continued for 47 minutes.

MISSION

**Contain risk, preserve evidence, identify system causes and produce a verified CAPA plan.**

Roles: incident lead · production manager · quality/reliability representative


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ACTIVITY 07

# Scenario questions

## DISCUSS / DECIDE

- Which system weakness allowed the alarm to be ignored?
- What evidence must be preserved before resetting the workstation?
- How will you prove the CAPA was effective?

## QUALITY OF REASONING

- Work from the evidence provided.
- State assumptions and missing evidence.
- Use current controlled limits and methods.
- Challenge the first explanation before concluding.


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ACTIVITY 07 · VERIFY

# Evidence and acceptance gate

## EVIDENCE TO SUBMIT

- Containment record
- Timeline and causal analysis
- CAPA plan
- Effectiveness-verification plan

## ACCEPTANCE CRITERIA

✓ Exposure window and product scope are defined

✓ Causes extend beyond individual blame

✓ Actions address root and detection causes

✓ Effectiveness has measurable evidence


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ACTIVITY 08 · K1–K4 · A1–A3

# Management Capstone: Build the Apex ESD Control Plan

Apex Microsystems is opening a mixed lab/factory line for sensitive industrial controllers. Management needs an auditable ESD control plan before customer qualification, covering controls, verification, competence, incidents and improvement.

**MISSION**

**Integrate the complete course into a management-ready ESD control plan and defend it under assessment-style questioning.**

Roles: ESD programme manager · operations manager · quality manager · customer assessor


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ACTIVITY 08

# Scenario questions

## DISCUSS / DECIDE

- How does the plan demonstrate A1, A2 and A3?
- Which records prove the control programme remains effective?
- What management decision is triggered by deteriorating leading indicators?

## QUALITY OF REASONING

- Work from the evidence provided.
- State assumptions and missing evidence.
- Use current controlled limits and methods.
- Challenge the first explanation before concluding.


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ACTIVITY 08 · VERIFY

# Evidence and acceptance gate

## EVIDENCE TO SUBMIT

- ESD control plan
- Qualification and verification matrices
- Competency matrix
- Metrics dashboard and 90-day roadmap

## ACCEPTANCE CRITERIA

✓ All K and A criteria are traceable

✓ Controls have owners and evidence

✓ Failure response and CAPA are closed-loop

✓ Plan uses current controlled standards for exact requirements


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RETRIEVAL PRACTICE

# Module 4 recap

1

## Risk

Competence · behaviour · reporting · root-cause analysis · CAPA · management ownership

2

## Knowledge

K4 · A2 · A3

3

## Evidence

Explain the decision and retain traceable records.

4

## Next

Carry the controls into the following module.


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COURSE SUMMARY

# What You Achieved

1

## Recognise

Explain charge generation, induction, discharge models and damage.

2

## Control

Design an EPA and select grounding, ionisation, shielding and packaging.

3

## Verify

Choose instruments, evaluate records and audit programme effectiveness.

4

## Improve

Investigate incidents, recommend CAPA and manage competence.


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# Courseware and Assessment on the LMS

1. Open the LMS portal
2. Select this course record
3. Download current courseware
4. Complete WA and Case Study
5. Upload completed answers

Open the LMS portal


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END-OF-COURSE REMINDER

# Assessment

## WA

- Written Assessment (SAQ): 1 hour.
- Four open-ended questions covering K1–K4.

## CASE STUDY

- Case Study (CS): 1 hour.
- Two open-ended tasks covering A1–A3.


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# Assessment Flow · Final Reminder

1. TRAQOM digital attendance
2. Assessment digital attendance
3. Sit WA then Case Study
4. Submit answers on the LMS
5. Sign Assessment Summary Record


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# Digital Attendance (Mandatory)

1. **TRAQOM**
   Scan the live assessment attendance QR code.

2. **Verify**
   Confirm successful submission before assessment.

3. **Privacy**
   Use your own details and device.

4. **Support**
   Ask the trainer if the live code does not open.


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THANK YOU

# ESD control is a system of engineered paths, disciplined behaviour and verified evidence.

Thank you · enquiry@tertiaryinfotech.com · +65 6100 0613


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